



A modified k - ω turbulence model for improved predictions of neutral atmospheric boundary layer flows

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ABSTRACT

A modified k - ω turbulence model is developed for neutral atmospheric boundary layer flows by introducing additional source terms to the equations of the standard model to remove its inconsistency with the inlet conditions. Rough wall functions based on aerodynamic roughness are adopted to overcome the limitations of the standard wall functions that are based on sand grain roughness. Moreover, the diagnostic and prognostic approaches are considered when simulating the flow. The diagnostic approach is approximate, relying on satisfying mass conservation by solving an optimization problem, whereas the prognostic approach resolves the Navier-Stokes equations with a turbulence model. The performances of the developed modified k - ω model, an existing modified k - ϵ model, and the standard SST k - ω model are compared. To calculate the concentration field, the Eulerian approach which solves a concentration transport equation, and the Lagrangian approach which adopts the particle method with core radius spreading for computing diffusion are used. Results generated with the diagnostic approach demonstrate its unsuitability to calculate wind flows over complex geometries. On the other hand, the prognostic approach, with the three mentioned turbulence models, was successful in reproducing the flow data of CEDVAL A1 and B1 experiments in domains composed of one building and an array of buildings. In addition, a good agreement with the concentration measurements is obtained when using the newly modified k - ω and the SST k - ω models to account for turbulence, with the Eulerian method outperforming the Lagrangian dispersion approach.

1. Introduction

Air pollution in dense urban areas is a threat to human health, quality of life, and sustainability. Accurate predictions of the wind flow field and the dispersion of air pollutants enable assessment of mitigation strategies, effective air quality management, and drive policies towards a more sustainable environment. Modeling of air pollution transport in urban environments is challenging due to the complexity of the flow field that arises from the interaction of the upstream atmospheric flow with the buildings. Laboratory experiments for studying pollution dispersion in urban areas are costly [1,2] and difficult to apply in practice. With the rapid increase in computational power, numerical models have become standard tools to simulate flow and pollutant dispersion in complex urban environments.

Modeling approaches to study pollution dispersion depend on the complexity of the problem. For example, Gaussian-type dispersion such as the California Puff Model (CALPUFF) [3] and the American Meteorological Society/Environmental Protection Agency Regulatory Model

(AERMOD) [4] are suitable for predicting concentration fields far from obstacles. In contrast, for domains with complex geometries, such as urban environments, Computational Fluid Dynamics (CFD) is increasingly used for studying the dispersion of pollutants. This approach, which is based on solving the Navier Stokes equations coupled with the problem descriptive equations, is known as the prognostic wind field modeling for its ability to forecast the time evolution of the atmospheric flow field. A cost effective yet less accurate alternative is the diagnostic wind field modeling, which downscales the larger scale flow field by imposing the conservation of mass on an interpolated meteorological wind field. The velocity field generated by the prognostic, or the diagnostic model, can then be used to predict the concentration field by solving the transport equation on a grid or using particles.

Mass-consistent diagnostic models impose continuity by solving an optimization equation that is based on variational analysis [5-7]. Several mass-consistent diagnostic models have been developed to simulate wind fields and the pollutants transport. The differences among these models are related to the way the initial wind field is constructed, the approach for calculating the equation weights, and the method for

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Nomenclature

f_c	Coriolis parameter
C_1, C_2	constants in turbulent kinetic energy inlet profile
C_D	drag coefficient
$C_{measure}$	measured concentration
C_s	shape and spacing roughness constant
C_{source}	source concentration
$C_{\epsilon 1}, C_{\epsilon 2}, C_\mu$	turbulence model parameters
D_t	eddy diffusivity
G_k	turbulence production
K_s	sand grain roughness
Q_{source}	flow rate of the tracer gas
S_c	source in c equation
S_ϵ	source in ϵ equation
S_ω	source in ω equation
Sc_t	Schmidt number
u^*	frictional velocity
U_h	velocity at reference height
u_o	measured wind velocity
$\hat{u}$	interpolated wind velocity
Z_0	altitude of measured velocity
z_0	surface roughness
c	concentration
C	normalized concentration
D	effective diffusion coefficient
E	wall treatment integration constant
h	reference height
k	turbulence kinetic energy
M	mass of Lagrangian particle
$\mathbf{n}$	surface normal unit vector
p	exponent obtained from the atmospheric stability

r	conditions; also, pressure
$\mathbf{u}$	distance between interpolated and measured velocity
u, v, w	velocity vector
$\mathbf{x}$	velocity components in x, y, and z direction
Z	position vector
α_1, α_2	altitude of interpolated velocity
β^*, β	Symbols
μ_{eff}	Gauss precision moduli
μ_t	turbulence model parameters
ν_t	effective viscosity
$\sigma_{(t)}$	turbulent dynamic viscosity
$\sigma_k, \sigma_\epsilon, \sigma_\omega$	turbulent kinematic viscosity
ΔB	core radius
α	turbulence model parameters
δ	departure from smooth conditions
κ	blending smoothness parameter
κ	deviation from the ABL conditions
λ	diffusion tensor
μ	Von Karman constant
ρ	Lagrange multiplier
τ	laminar viscosity
φ	density
ω	stress tensor
ϵ	the model parameter to be blended
	turbulence specific dissipation rate
	turbulence dissipation rate

Subscripts

ABL	values for the atmospheric boundary layer region
BIA	values for the building influence area region
g	value at wall boundary cell centroid
p	refers to particle

solving the equations. The divergence-free wind flow model MATHEW was developed by Sherman [8] to provide the wind field for the ADPIC pollutant transport model [9] and was applied for the geysers drainage flow region in California [10], for a region in Colorado [11], and for the Apennine Mountains during the Lago Brasimone experimental campaign [12]. The NUATMOS mass consistent model [13] was developed and used to simulate tracer transport of the Cinder Cone Butte experiment [14]. The mass consistent model NOABL [15] was used for simulating wind flow over Blashaval Hill [16,17]. It was tested also with the COMPLEX wind model [18] in three regions of the UK, which resulted in some modifications and led to the development of a new MC-3 wind model [19]. Mass-consistent wind models were used for downscaling purposes, for increasing the resolution of wind fields generated by Global Circulation Models, and for regional-scale dispersion meteorological pre-processing [20,21]. Another application for diagnostic wind field models was an instantaneous simulation for pollution control networks in an industrial plant [22].

The above diagnostic models do not yield sufficiently accurate velocity fields over complex geometries, especially when the flow is turbulent, which is mostly the case for wind. To this end, prognostic models are used. The prognostic approach for solving turbulent flows can be divided broadly into the direct numerical simulation (DNS), large-eddy simulation (LES), and the Reynolds-Averaged Navier-Stokes (RANS) closure. DNS incurs huge computational costs, LES on the other hand, has a high but acceptable computational cost for simple turbulent flow and dispersion problems [23,24], while the RANS approach is currently the most feasible for industrial-size applications [25,26]. Several RANS turbulence closures were developed and used for wind and dispersion modeling in urban areas. They include the standard $k-\epsilon$ model [27], the realizable $k-\epsilon$ model [28], the renormalization group $k-\epsilon$ model (RNG)

[29], the standard $k-\omega$ model [30–32] and the $k-\omega$ shear stress transport model (SST $k-\omega$) [33]. Currently, none of these models is capable of accurately predicting solutions over the whole domain of study [34], and each of them is suitable for a particular class of problems [35]. As such, there is a continuous effort to improve these models and overcome their drawbacks. Thus, the choice of turbulence model for air flow in complex urban environments is still not well explained theoretically [36].

The Atmospheric Boundary Layer (ABL) approximation assumes a fully developed and horizontally homogeneous flow at the inlet. In line with this assumption, Richards and Hoxey [37] proposed boundary conditions that are consistent with the $k-\epsilon$ turbulence closure. Generally, an ABL domain can be divided into three parts, which are the upstream and downstream regions where buildings are modeled implicitly using rough wall functions, and the central region where buildings are modeled explicitly using their real geometries. Usually, roughness is treated using wall functions with sand-grain roughness [38], which are modified versions of the standard wall functions [39]. However, Blocken et al. [40] explained that the law of the wall for rough surfaces when modeling buildings implicitly is not consistent with the inlet profiles proposed by Richards and Hoxey [37]. In addition, Richards and Hoxey [37] proposed a uniform inlet profile for turbulent kinetic energy, which does not agree with experimental measurements. To address this problem, Yang et al. [41] derived analytically a variable inlet profile for turbulent kinetic energy that satisfies measurements, but without fully studying its impact on the used model. To ensure consistency with the variable k profile, Parente et al. [42] modified the $k-\epsilon$ model and proposed a new implementation of wall functions based on aerodynamic roughness to overcome the limitations of the sand-grain rough wall functions [40].

In this study, a numerical tool capable of simulating wind flow and dispersion of a non-reacting pollutant using the diagnostic and prognostic approaches is developed. In the diagnostic model, an interpolated velocity field is adjusted to satisfy mass conservation. In the prognostic model, the SIMPLEC algorithm is used with several turbulent closure schemes. The measures proposed by Richards and Hoxey, Yang et al., and Parente et al. [37,41,42] to obtain equilibrium between the k - ϵ turbulence model, inlet profiles, and wall functions are implemented and tested. Also, modifications to the k - ω turbulence model are proposed and compared with the modified k - ϵ model. Two approaches for modeling pollutant dispersion are used. The first solves the concentration transport equation in a Eulerian framework, while the second employs Lagrangian tracking of particles that models diffusion by core spreading. Wind flow models are validated using several test cases including CEDVAL A1-1 and CEDVAL B1 wind tunnel experiments, while dispersion models are validated using the CEDVAL A1-5 experiment [43].

2. Implementation and governing equations

2.1. Mesh generation

The Eulerian forms of the conservation equations are discretized on unstructured grid systems in the context of a collocated, cell-centered, and pressure-based finite volume method [45,46]. For the configurations considered in this work, the “blockMesh” OpenFOAM® utility is used to create hexahedral meshes for the cases of rectangular geometries, and the “SnappyHexMesh” OpenFOAM® utility for cases with curved geometries [44,45]. A code was developed to automate the creation of OpenFOAM® “blockMeshDict” and “SnappyHexMeshDict” dictionaries based on simple user specifications.

2.2. Diagnostic wind field model

The diagnostic model is a fast but approximate model that interpolates the observed coarsely spaced wind data onto the fine grid and then adjusts the interpolated field to meet the no divergence condition while minimizing the changes. Measured data are interpolated in two steps. First, interpolation is conducted in the vertical direction using the power law scheme (equation (1)), then in the horizontal directions using the nearest few measurements weighted inversely by the square of the distance (equation (2)).

$$\hat{u} = u_o \left(\frac{Z}{Z_0} \right)^p, \quad (1)$$

$$\hat{u} = \frac{\sum_{i=1}^n \frac{\hat{u}_i}{r_i^2}}{\sum_{i=1}^n \frac{1}{r_i^2}}. \quad (2)$$

In equation (1), u_o and $\hat{u}$ are the measured and the interpolated velocities respectively, Z_0 and Z are the altitudes of the measured and the interpolated velocities, and p is an exponent obtained from the atmospheric stability conditions [46] or a least-squares fit of data. In this work, p is calculated from a least-squares fit of the inlet velocity profile.

The interpolation is not straightforward for an unstructured grid. To simplify it, a structured grid is constructed over which interpolation is initially conducted and then transformed to the unstructured grid elements. The vertical interpolation is first done for the points (cell centroids) that are vertically aligned with measurements according to equation (1). Then, horizontal interpolation for the remaining points is performed using the values of the vertically interpolated points (equation (2)).

Finding the mass-conserving wind field, while minimizing the change from the initial field, is done by solving an anisotropic diffusion equation for a Lagrange multiplier λ (equation (3)). The predicted λ field is then used to update the velocity components according to the

Euler–Lagrange equations (equation (4)) [5,8].

$$\nabla \bullet [\nabla(\kappa\lambda)] = -2\alpha_1^2 \nabla \bullet \hat{\mathbf{u}}, \kappa = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & \frac{\alpha_1^2}{\alpha_2^2} \end{bmatrix}, \quad (3)$$

$$u = \hat{u} + \frac{1}{2\alpha_1^2} \frac{\partial \lambda}{\partial x}, v = \hat{v} + \frac{1}{2\alpha_1^2} \frac{\partial \lambda}{\partial y}, w = \hat{w} + \frac{1}{2\alpha_2^2} \frac{\partial \lambda}{\partial z}. \quad (4)$$

In equation (3), κ is the diffusion tensor and α_1 and α_2 are the Gauss precision moduli, which act as weights for adjusting the velocity components. Ratto et al. [47] defined a variable $\alpha = \alpha_1/\alpha_2$ as the “empirical stability parameter”. Their analysis indicated that the adjustment dominates in the vertical direction when $\alpha \gg 1$ and in the horizontal direction when $\alpha \ll 1$. Sherman [8] suggests that α can be chosen by relating it to the expected value of w/u . Two boundary conditions are used to solve the elliptic optimization equation (3) with the diagnostic approach. These are the closed (equation (5)) and the flow-through (equation (6)) boundary conditions. The closed boundary condition is applied for terrain and buildings while the open or flow through boundary condition is applied at all other boundaries.

$$\nabla \lambda \bullet \mathbf{n} = 0, \mathbf{u} = 0. \quad (5)$$

$$\lambda = 0. \quad (6)$$

2.3. Prognostic wind field model

The used methodology follows a pressure-based solution procedure and adopts a SIMPLE-like algorithm [48,49] to solve the steady incompressible continuity and momentum equations given by

$$\nabla \bullet \mathbf{u} = 0, \quad (7)$$

$$\nabla \bullet (\rho \mathbf{u} \mathbf{u}) = \nabla \bullet \tau - \nabla p, \quad (8)$$

where τ is the stress tensor and p the pressure. Since a RANS approach is adopted, the stress tensor is computed as

$$\tau = \mu_{eff} (\nabla \mathbf{u} + \nabla \mathbf{u}^T) - \frac{2}{3} \rho k \mathbf{I}. \quad (9)$$

The effective viscosity μ_{eff} is calculated as

$$\mu_{eff} = \mu + \mu_t, \quad (10)$$

where μ and μ_t are the laminar and turbulent viscosity, respectively. The k - ϵ and k - ω two-equation turbulence models are used in this work to calculate the turbulent viscosity. To overcome the limitations and inaccuracies of using these models in their standard forms, we use the modified k - ϵ model presented in Ref. [39] and propose similar modifications to the standard k - ω model. Results are also compared with those predicted by the SST k - ω model. Further, roughness is modeled using the rough wall functions reported in Refs. [42,50] that are suitable for ABL simulations. These modifications are detailed next.

2.3.1. The modified k - ϵ turbulence model

The k - ϵ turbulence model [39] involves solving the turbulent kinetic energy (k) and dissipation rate (ϵ) equations. Neglecting laminar viscosity and buoyancy effects, the steady-state forms of these equations are given by

$$\nabla \bullet (\rho \mathbf{u} k) = \nabla \bullet \left(\frac{\mu_t}{\sigma_k} \nabla k \right) + G_k - \rho \epsilon, \quad (11)$$

$$\nabla \bullet (\rho \mathbf{u} \epsilon) = \nabla \bullet \left(\frac{\mu_t}{\sigma_\epsilon} \nabla \epsilon \right) + C_{\epsilon 1} G_k \frac{\epsilon}{k} - C_{\epsilon 2} \rho \frac{\epsilon^2}{k}, \quad (12)$$

where σ_k , σ_ε , $C_{\varepsilon 1}$, and $C_{\varepsilon 2}$ are model parameters. As discussed by Richards and Hoxey [37], for steady incompressible modeling of the atmospheric boundary layer, the assumption of a homogenous flow implies that the pressure is constant, the vertical velocity is zero, and the shear stress is constant. Thus, the momentum, turbulent kinetic energy, and turbulent dissipation rate equations reduce to

$$\mu_t \frac{\partial u}{\partial z} = \tau = \rho u^{*2}, \quad (13)$$

$$\frac{\partial}{\partial z} \left(\frac{\mu_t}{\sigma_k} \frac{\partial k}{\partial z} \right) + G_k - \rho \varepsilon = 0, \quad (14)$$

$$\frac{\partial}{\partial z} \left(\frac{\mu_t}{\sigma_\varepsilon} \frac{\partial \varepsilon}{\partial z} \right) + C_{\varepsilon 1} G_k \frac{\varepsilon}{k} - C_{\varepsilon 2} \rho \frac{\varepsilon^2}{k} = 0. \quad (15)$$

The production of turbulent kinetic energy is $G_k = \mu_t (\partial u / \partial z)^2$, and the turbulent viscosity is $\mu_t = \rho C_\mu k^2 / \varepsilon$, where C_μ is a model parameter assigned the value 0.09 in the standard model. ABL inlet profiles are generally fully developed, and to ensure the consistency between them and the k - ε turbulence model, Richards and Hoxey [37] proposed the following boundary conditions satisfying equations (13)–(15):

$$u^* = \frac{\kappa U_h}{\ln \left(\frac{h+z_0}{z_0} \right)}, u = \frac{u^*}{\kappa} \ln \left(\frac{z+z_0}{z_0} \right), k = \frac{u^{*2}}{\sqrt{C_\mu}}, \varepsilon = \frac{u^{*3}}{\kappa(z+z_0)}, \quad (16)$$

where z_0 is the surface roughness height, κ is the Von Karman constant, u^* is the frictional velocity, and U_h is the known velocity at reference height h . Similarly, at walls

$$u_g^* = \frac{\kappa U_g}{\ln \left(\frac{z_g+z_0}{z_0} \right)}, \tau_g = \rho u_g^{*2}, k_g = \frac{u_g^{*2}}{\sqrt{C_\mu}}, \varepsilon_g = \frac{\sqrt{C_\mu} k u_g^*}{\kappa(z_g+z_0)}, \quad (17)$$

where U_g , k_g and ε_g are velocity, turbulent kinetic energy, and turbulent dissipation rate values at wall boundary cell centroids. At the top surface, constant shear stress $\tau = \rho u^{*2}$ is recommended. The suggested inlet profile proposed in equation (16) automatically satisfies equation (14) but to satisfy equation (15), the turbulent dissipation Prandtl number should be computed as

$$\sigma_\varepsilon = \frac{\kappa^2}{(C_{\varepsilon 2} - C_{\varepsilon 1}) \sqrt{C_\mu}}. \quad (18)$$

The drawback of Richards and Hoxey's [37] inlet profiles is that the constant k inlet profile equation (14) is not in agreement with experimental data. To overcome this limitation, Yang et al. [41] assumed equilibrium between turbulence production and dissipation requiring the first term in equation (14) to be zero, which results in the following profile for the turbulent kinetic energy:

$$k(z) = \sqrt{C_1 \ln(z+z_0) + C_2}, \quad (19)$$

where C_1 and C_2 are constants deduced from measurements. Parente et al. [42] followed a different route to modify the k - ε model enabling the use of non-constant k inlet profiles. In their approach, the dissipation rate ε and C_μ are expressed as

$$\varepsilon(z) = \sqrt{C_\mu} k \frac{du}{dz}, C_\mu(z) = \frac{u^{*4}}{k(z)^2}. \quad (20)$$

Then, by substituting $C_\mu(z)$ and $k(z)$ in equation (14), the following additional source term appears:

$$S_k(z) = \frac{\rho u^* \kappa}{\sigma_k} \frac{\partial}{\partial z} \left((z+z_0) \frac{\partial k}{\partial z} \right). \quad (21)$$

This source term should be added to the turbulent kinetic energy equation (14) to ensure equilibrium. For non-constant C_μ , instead of using equation (18) to calculate σ_ε , the constant σ_ε value is maintained

while adding the following source term to the ε equation (15):

$$S_\varepsilon(z) = \frac{\rho u^{*4}}{(z+z_0)^2} \left(\frac{(C_{\varepsilon 2} - C_{\varepsilon 1}) \sqrt{C_\mu}}{\kappa^2} - \frac{1}{\sigma_\varepsilon} \right). \quad (22)$$

Further, Parente et al. [50] derived a new consistent inlet profile for turbulent kinetic energy based on the assumption of local equilibrium between dissipation and production of turbulence, as

$$k(z) = C_1 \ln(z+z_0) + C_2. \quad (23)$$

The difference between equations (19) and (23) is that the first one needs the additional source term (equation (21)) in the turbulent kinetic energy equation (equation (14)) to force equilibrium while the second one is derived based on equilibrium condition not requiring any additional source. Both Yang et al. [41] and Parente et al. [50] proposed accurate profiles that require experimental data to fit the parameters C_1 and C_2 in equations (19) and (23). If experimental data is not available to calculate C_1 and C_2 , the following semi-empirical relation provided by Brost and Wyngaard [51] can be used:

$$k(z) = \frac{u^{*2}}{2} \left(8.7 - 6 \frac{z}{h} \right), \quad (24)$$

where h is the ABL height obtained from the Bechmann [52] relation as

$$\frac{hf_c}{u^{*2}} \approx 0.33, \quad (25)$$

with the Coriolis parameter f_c set equal to 10^{-4} . The use of this profile requires adding the source term given in equation (21) to the turbulent kinetic energy equation to force equilibrium.

2.3.2. The modified k - ω turbulence model

The k - ω turbulence model [53] requires solving the turbulence kinetic energy (k) and specific dissipation rate (ω) equations. Neglecting laminar viscosity and buoyancy effects, the steady-state forms of these equations are given by

$$\nabla \bullet (\rho \mathbf{u} k) = \nabla \bullet \left(\frac{\mu_t}{\sigma_k} \nabla k \right) + G_k - \beta^* \rho \omega k + S_k, \quad (26)$$

$$\nabla \bullet (\rho \mathbf{u} \omega) = \nabla \bullet \left(\frac{\mu_t}{\sigma_\omega} \nabla \omega \right) + \frac{\alpha \omega}{k} G_k - \beta \rho \omega^2 + S_\omega, \quad (27)$$

where σ_k , σ_ω , β^* , and β are model parameters. Following Parente et al.'s approach for modifying the k - ε model [50], we derive new source terms for the k and ω equations leading to a modified model appropriate for simulating ABL flows. First, the ω inlet profile is derived based on the velocity inlet profile proposed by Richards and Hoxey [37]. To relate ω and velocity, the turbulent viscosity $\mu_t = \rho k / \omega$ is substituted in equation (13) to yield

$$\omega = \rho \frac{k}{\tau} \frac{\partial u}{\partial z}. \quad (28)$$

For the case of a homogenous ABL flow, turbulence generation ($\tau \partial u / \partial z$) and dissipation ($\beta^* \rho k \omega$) are in equilibrium, so that

$$\tau \frac{\partial u}{\partial z} = \beta^* \rho k \omega \Rightarrow \beta^* = \frac{u^{*4}}{k(z)^2}. \quad (29)$$

Substituting the expressions for k and τ from equations (29) and (13) in equation (28) respectively, and using equation (16) to calculate $\partial u / \partial z$, the ω profile at the inlet becomes

$$\omega = \frac{u^*}{\sqrt{\beta^*} \kappa} \frac{1}{z+z_0}. \quad (30)$$

The inlet profile should be consistent with the ω transport equation, which means that the inlet ω profile should satisfy the ω equation. For two-dimensional, steady, incompressible, homogenous ABL flow, the ω

equation (27) reduces to

$$\frac{\partial}{\partial z} \left[\left(\frac{\mu_t}{\sigma_\omega} \right) \frac{\partial \omega}{\partial z} \right] + \alpha \frac{\omega}{k} \tau \frac{\partial u}{\partial z} - \beta \rho \omega^2 + S_\omega = 0. \quad (31)$$

The terms of equation (31) are given by

$$\left\{ \begin{array}{l} \frac{\partial}{\partial z} \left[\left(\frac{\mu_t}{\sigma_\omega} \right) \frac{\partial \omega}{\partial z} \right] = \frac{\rho u^{*2}}{\sigma_\omega \sqrt{\beta^*}} \frac{1}{(z + z_0)^2} \\ \alpha \frac{\omega}{k} \tau \frac{\partial u}{\partial z} = \frac{\alpha \rho u^{*2}}{\kappa^2 (z + z_0)^2} \\ \beta \rho \omega^2 = \frac{\beta \rho u^{*2}}{\beta^* \kappa^2 (z + z_0)^2} \end{array} \right. \quad (32)$$

Upon substituting the terms of equation (32) in equation (31) an extra source term for ω appears as

$$S_\omega = \frac{\rho u^{*2}}{(z + z_0)^2 \sqrt{\beta^*}} \left[\frac{\beta - \alpha \beta^*}{\sqrt{\beta^*} \kappa^2} - \frac{1}{\sigma_\omega} \right]. \quad (33)$$

The source term in equation (33) should be added to the ω equation to insure consistency between the inlet profile and model. For the k equation, the same source term as for the k - ϵ model given by equation (21) is obtained.

At the top of the domain, a specified flux boundary condition for ω is used and is given by

$$Flux T_\omega = \frac{\mu_t}{\sigma_\omega} \frac{\partial \omega}{\partial z}. \quad (34)$$

Using the ω inlet profile, expressed in equation (30), the flux becomes

$$Flux T_\omega = \frac{\rho u^{*2}}{\sqrt{\beta^*} \sigma_\omega (z + z_0)}. \quad (35)$$

2.3.3. The SST k - ω turbulence model

Like the k - ω model, the SST k - ω turbulence model [54] involves solving the turbulent kinetic energy (k) and the specific dissipation rate (ω) equations given by

$$\nabla \bullet (\rho \mathbf{u} k) = \nabla \bullet [(\mu + \sigma_k \mu_t) \nabla k] + P_k - \beta^* \rho \omega k, \quad (36)$$

$$\nabla \bullet (\rho \mathbf{u} \omega) = \nabla \bullet [(\mu + \sigma_\omega \mu_t) \nabla \omega] + \alpha \rho \frac{P_k}{\mu_t} - \beta \rho \omega^2 + 2(1 - F_1) \frac{\rho \sigma_{\omega 2}}{\omega} \nabla k \bullet \nabla \omega. \quad (37)$$

Description of the various terms and model constants in equations (36) and (37) can be found in Ref. [54]. No modifications are used with this model.

2.3.4. Treatment at a wall

As mentioned before, ABL domains are generally divided into the central region where objects are modeled explicitly and upstream and downstream regions where objects are modeled implicitly by including their effect on flow as wall roughness. For rough surfaces, the following modified wall functions based on sand grain roughness are usually used:

$$U^+ = \frac{1}{\kappa} \ln(E y^+) - \Delta B, \quad (38)$$

where $U^+ = u_p u^* / (\tau_w \rho)$, $y^+ = \rho u^* y_p / \mu$, E is an integration constant, and u^* is the frictional velocity. The term ΔB represents the departure from smooth conditions, and it can be calculated according to Cebeci and Bradshaw [38], depending on sand grain roughness K_s^+ , as

$$\left\{ \begin{array}{l} \Delta B = 0 \quad K_s^+ < 2.25 \\ \Delta B = \frac{1}{\kappa} \log \left(\frac{K_s^+ - 2.25}{87.75} + C_s K_s^+ \right) \sin(0.4258 [\log K_s^+ - 0.811]) \quad 2.25 < K_s^+ < 90 \\ \Delta B = \frac{1}{\kappa} \log(1 + C_s K_s^+) \quad K_s^+ > 90 \end{array} \right. \quad (39)$$

However, Blocken et al. [40] showed that using K_s -type wall functions cannot achieve a horizontally homogeneous ABL flow without some errors and proposed some remedial measures to rectify it. Alternatively, Parente et al. [42] developed a new wall function based on aerodynamic length to overcome K_s -type wall function limitations, which is a reformulation of the one reported in Ref. [37]. The difference is in calculating $G_k = \tau_w^2 / \rho \kappa C_\mu^{0.25} K_p^{0.5} (z_p + z_0)$ at cell centroid instead of integrating it over the first cell height to avoid peak production of k at the wall [55]. The new wall function preserves the universal form $u = (u^* / \kappa) \ln(E' z'^+)$, where E' and z'^+ are defined as

$$E' = \begin{cases} E \text{ smooth} \\ \frac{\nu}{z_0 u^*} \text{ rough} \end{cases} \quad z'^+ = \begin{cases} \frac{z_p u^*}{\nu} \text{ smooth} \\ \frac{(z_p + z_0) u^*}{\nu} \text{ rough} \end{cases} \quad (40)$$

The frictional velocity in equation (40) is variable and is calculated as $u^* = C_\mu^{0.25} k_p^{0.5}$. The new wall function formulation improves the standard approach while preserving its flexibility.

2.3.5. Blending the models according to buildings' influence area

The modified k - ϵ and k - ω formulations are valid when the ABL is undisturbed. Thus, in the building influence area where buildings are modeled explicitly using their real shapes, the standard models should be used. To detect the building influence area automatically, and to provide a smooth transition and blending between the implicit (where buildings are modeled via roughness) and explicit regions, a method based on the local deviation from the undisturbed ABL conditions is used [50]. The deviation from ABL conditions (δ) can be calculated by pure blending based on u (equation (41)) or based on k (equation (42)) or by hybrid blending based on both u and k (equation (43)) [56].

$$\delta_u = \min \left[\left| \frac{u - u_{ABL}}{u_{ABL}} \right|, 1 \right], \quad (41)$$

$$\delta_k = \min \left[\left| \frac{k - k_{ABL}}{k_{ABL}} \right|, 1 \right], \quad (42)$$

$$\delta_h = \max[\delta_u, \delta_k]. \quad (43)$$

The additional source terms of the k , ϵ , and ω equations and the C_μ and β^* parameters are then weighted as a function of δ to allow a smooth transition between explicit and implicit regions. Two formulations for the transition between the flow regions are available, based on a polynomial (equation (44)) and a sinusoidal function (equation (45)).

$$\varphi = \varphi_{BIA} + (1 - \delta^\alpha)(\varphi_{ABL} - \varphi_{BIA}), \quad (44)$$

$$\left\{ \begin{array}{l} \varphi = \varphi_{BIA} + (\varphi_{ABL} - \varphi_{BIA}) [1 - 0.5(1 + \sin(\delta^*))]^\alpha \\ \delta^* = \pi \max(\delta - 0.5, -0.5) \end{array} \right. \quad (45)$$

In the above equations φ is the model parameter to be modified, and φ_{BIA} and φ_{ABL} are the values of φ in the building influence area and the ABL regions, respectively. The parameter α controls the smoothness of the blending, and more details for choosing it can be found in Ref. [56].

The full set of inlet conditions and source terms are summarized in

Table 1a

Summary of inlet conditions and turbulence model additional source terms.

Variable	Equation	Additional Source
$u(z)$	Eq. (16)	0
$k(z)$	Eq. (16) or Eq. (23)	0
	Eq. (19) or Eq. (24)	Eq. (21)
$\epsilon(z)$	Eq. (16)	Eq. (22)
$\omega(z)$	Eq. (30)	Eq. (33)

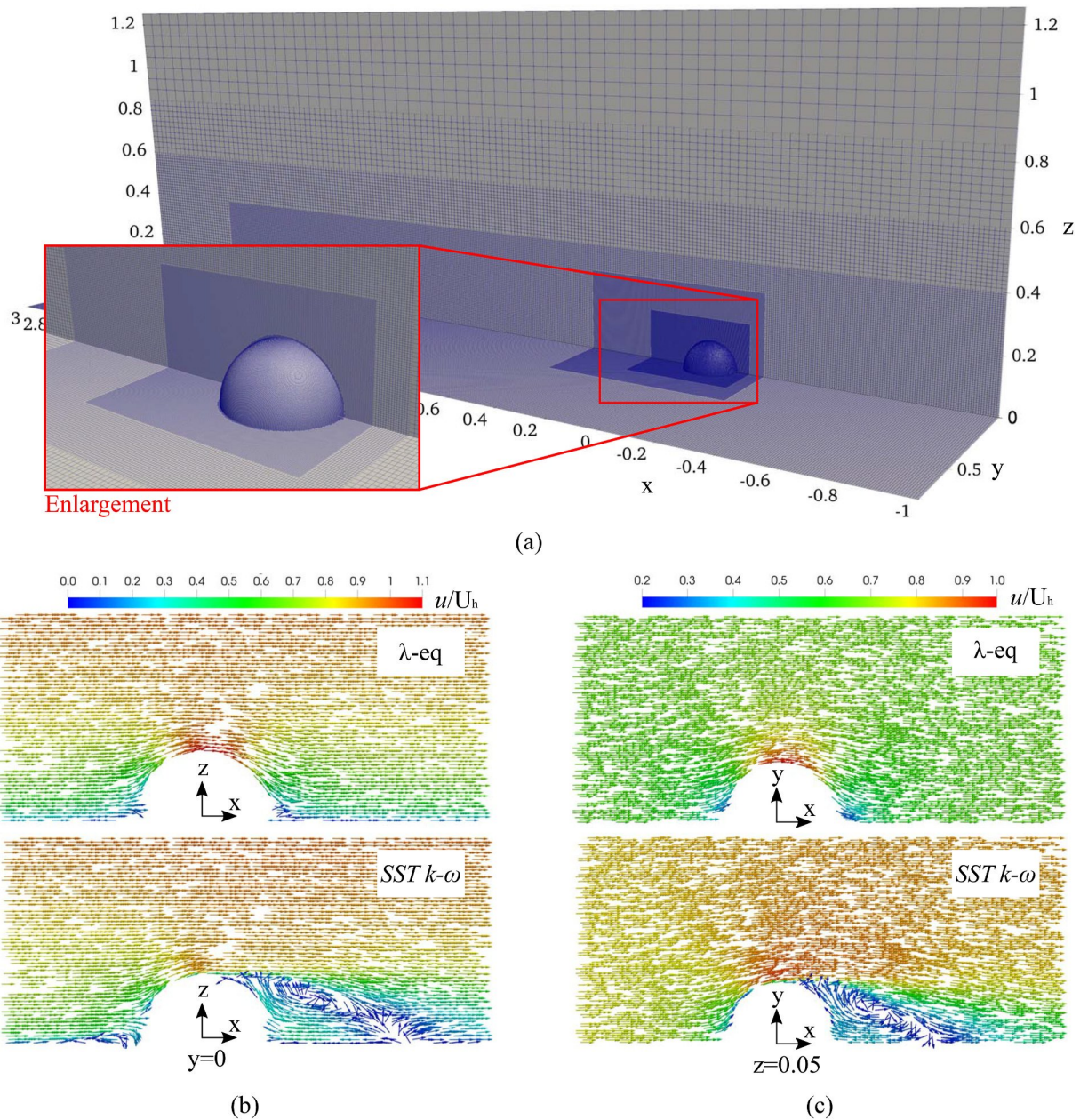


Figure 1. (a) Physical domain for the flow over a hemispherical obstacle; (b) Comparison of diagnostic (λ -eq) and prognostic ($SST k-\omega$) velocity fields around the hemispherical obstacle in the vertical plane $y = 0$; (c) Comparison of diagnostic (λ -eq) and prognostic ($SST k-\omega$) velocity fields around the hemispherical obstacle in the vertical plane $z = 0.05$ (x, y , and z in m; velocity in m/s).

Table 1a. Hybrid or pure blending can be used to calculate δ , and either equation (44) or equation (45) can be used to transit between implicit and explicit regions. Equation (40) can be used for wall functions instead of the standard rough wall functions.

2.4. Dispersion models

In the Lagrangian model, pollutants or gases are represented as a group of particles to simulate their distribution in the domain. The dispersion of particles and the concentration field are calculated by dividing the effects of advection and diffusion into two separate passive steps. The position of each particle is found by solving the advection equation while the effect of the particle on the concentration field is computed by the diffusion equation.

The Lagrangian form of Newton's second law applied to a particle is integrated to get the advection equation (equation (46)).

$$m_p \frac{d\mathbf{u}_p}{dt} = \overbrace{\frac{1}{2} \rho_f C_D |\mathbf{u} - \mathbf{u}_p| (\mathbf{u} - \mathbf{u}_p) A_p}^{\text{drag}} + \overbrace{(\rho_p - \rho_s) \mathbf{g} V_p}^{\text{buoyancy}} + \mathbf{F}_{\text{others}}. \quad (46)$$

Then, the sequence of events for advecting particles is summarized as follows:

1. Calculate the Reynolds number R_e using $\mathbf{u}_p$ from the previous time step as

$$R_e = \frac{\rho_f d_p |\mathbf{u} - \mathbf{u}_p|}{\mu}. \quad (47)$$

2. Calculate the drag coefficient C_D [57] based on R_e as

$$C_D = \begin{cases} \frac{24}{R_e} (1 + 0.15 R_e^{0.687}) & R_e < 1000 \\ 0.44 & R_e > 1000 \end{cases} \quad (48)$$

3. Solve the force balance (equation (46)) to calculate u_p .
4. Update the particle position by solving the trajectory equation given by

$$\frac{dx_p}{dt} = u_p. \quad (49)$$

The force balance and trajectory equations are solved using the fourth order Runge-Kutta method.

An equation for particle diffusion is developed by solving Fick's law [58] using the "product rule" for the case of a particle spreading in all directions without being restricted by boundaries [59]. The equation describes how a particle of mass M deposited at a point (x_0, y_0, z_0) at time $t = 0$ affects the concentration at point (x, y, z) , and it can be written as

$$c(x, y, z, t) = \frac{M}{(4\pi t)^{\frac{3}{2}} (D_x D_y D_z)^{1/2}} \exp \left(-\frac{(x-x_0)^2}{4D_x t} - \frac{(y-y_0)^2}{4D_y t} - \frac{(z-z_0)^2}{4D_z t} \right), \quad (50)$$

where D_x, D_y and D_z are the diffusivities in x, y , and z directions. If the diffusivity is isotropic, the core-radius can be defined as $\sigma_{(t)}^2 = 4Dt$, and the equation becomes

$$c(x, y, z, t) = \frac{M}{\pi^{3/2} \sigma_{(t)}^3} \exp \left(-\frac{d^2}{\sigma_{(t)}^2} \right), \quad \sigma_{(t)} = \sqrt{\sigma_{(t-l)}^2 + 4Ddt}. \quad (51)$$

In the simulations, the concentration field is described by the number of particles, and the effect of all particles on each cell centroid is calculated by superposing the effects of particles according to

$$c(x, y, z, t) = \sum_{i=1}^{\# \text{ particles}} \frac{M_i}{\pi^{3/2} \sigma_{(i,t)}^3} \exp \left(-\frac{d_i^2}{\sigma_{(i,t)}^2} \right). \quad (52)$$

On the other hand, the concentration field in the Eulerian model is estimated by solving a concentration transport equation given by

$$\frac{\partial c}{\partial t} + \nabla \cdot (\mathbf{u}c) = \nabla \cdot [D \nabla c] + S_c, \quad (53)$$

where c is the concentration, $\mathbf{u}$ is the wind velocity, D is the effective diffusion coefficient, and S_c is a source term. The effective diffusivity D is the sum of the molecular and eddy diffusivities, with the eddy diffusivity defined as

$$D_t = \frac{\nu_t}{Sc_t}, \quad (54)$$

where Sc_t is the Schmidt number that varies between 0.2 and 1.3 [60], and ν_t is the kinematic turbulent viscosity.

3. Test cases

Different cases are simulated to evaluate the performance of the diagnostic and prognostic wind field models. The performance of the approximate diagnostic model is compared with that of the SST $k-\omega$ model in a domain with a hemispherical obstacle to measure its accuracy. To validate the prognostic wind field model, first, the CEDVAL A1 [43] wind tunnel experiment is simulated in the absence of obstacles to demonstrate the sustainability of the ABL velocity and turbulence inlet profiles. Then, the flow and dispersion of particles are simulated in the presence of a single building for the CEDVAL A1 test case. After that, the model is further validated by simulating the more challenging CEDVAL B1 test case dealing with an array of buildings.

For the solution of the anisotropic diffusion equation (Equation (3)),

the method developed by Darwish and Moukalled [61] is used. A second-order accurate, semi-implicit method is adopted for the discretization of the isotropic and anisotropic diffusion terms. In this approach, the diffusion flux is decomposed into two parts, one implicit and one explicit, with the decomposition following the procedure reported by Moukalled et al. [61,62]. For all problems reported in this work, the second order "MINMOD" high-resolution convection scheme is employed [48]. Pressure velocity coupling is resolved using the SIMPLEC algorithm [63,64]. For all test cases, a solution is considered converged when the rms residuals for all variables drop below 10^{-6} .

3.1. Flow over a hemispherical obstacle

The physical situation is schematically displayed in Fig. 1a. As depicted, a hemispherical obstacle of radius 0.1 m is positioned at the origin in a domain that is 4 m long, 1.5 m wide, and 1.25 m high. The inlet and outlet boundaries are located 1 m upstream and 3 m downstream of the origin, respectively. Moreover, the flow at the inlet is taken to be in the x -direction creating a symmetrical velocity profile with respect to the $y = 0$ plane and reducing the computational domain into half the physical domain.

As experimental data is unavailable, the problem is first solved via the prognostic approach using the SST $k-\omega$ turbulence model. For that purpose, the computational domain is covered by a mesh composed of 4,160,751 elements with denser clustering of grid points near walls. The grid system over the entire domain along a magnified region around the hemispherical obstacle is depicted in Fig. 1a. At the inlet, the velocity, turbulent kinetic energy, and specific dissipation rate profiles are specified and computed using equation (16) (to model a fully developed turbulent wind field) using the following parameters: $z_0 = 0.00075$ m, $U_h = 1$ m/s, and $h = 0.4$ m. A no-slip boundary condition is applied on walls, a specified pressure at the outlet, and symmetry elsewhere. The final solution resulted in y^+ values of less than 10 over the ground and less than 2 over the hemispherical obstacle.

Predicted values at twenty-seven locations are extracted from the solution and treated as observations to solve the problem using the diagnostic model. The locations of the observations are chosen randomly within the plane $z = 0.6$ m. It is possible to select more locations from the solution as observation points to help the method reach the correct solution. However, this does not correspond to real situations since measurements are usually available at a few locations in the domain [13]. As described earlier, the diagnostic solution is produced in two steps. First, an interpolated velocity field is constructed from the observations using equations (1) and (2). Then, equation (3) is solved subject to the boundary conditions described by equation (5) and equation (6) to predict the λ field and generated values are employed in equation (4) for updating the interpolated field to better satisfy continuity. The parameters in the problem are the exponent p needed in equation (1), and α_1 and α_2 used in equation (3). The exponent p is assigned the value of 0.19 as obtained from a least-squares fit of the logarithmic inlet velocity profile. The values for α_1 and α_2 are set at 1 and 2 leading to a value for α_1/α_2 of 0.5, which is within the interval of values [0.1,1] reported by Ross et al. [13]. Experimentation with different values of α_1/α_2 did not have noticeable effects on predictions. The velocity fields obtained by both models are compared in terms of velocity vectors (Fig. 1b and c) and profiles of velocity components.

3.2. CEDVAL A1 empty domain

Numerical simulations are carried out in an empty domain (CEDVAL A1 case) using the ABL $k-\epsilon$ and $k-\omega$ models, the standard and modified wall functions, and the different inlet profiles for the various variables listed in Table 1a. This was done to test the ability of the ABL $k-\epsilon$ and $k-\omega$ models to maintain the inlet profiles through empty domains. For boundary conditions, specified shear stress is applied on the top of the domain and a specified pressure at the outlet as recommended by

Table 1b
Velocity and turbulent kinetic energy model parameters.

u^*		0.378 ms^{-1}
z_0		0.00075 m
$C_s = Ez_0 / z_p$		$K_s = 30z_p$
Eq. (19) parameters	C_1, C_2	$-0.0628, 0.2186$
Eq. (23) parameters	C_1, C_2	$-0.0579, 0.4706$

Richards and Hoxey [37]. The two-dimensional empty domain is 4 m long and 1 m high. The computational grid is uniform in the horizontal direction and stretched in the vertical direction such that the height z_p of the first cell is 2.5 mm with a total number of 400×71 cells. The aerodynamic roughness z_0 is set at 0.00075 m, with the equivalent sand grain roughness K_s and roughness constant C_s calculated for standard rough wall functions based on the equations listed in Ref. [40]. The CEDVAL A1 case is well described in terms of velocity and turbulent kinetic energy measurements, which are used to calculate the model parameters displayed in Table 1b.

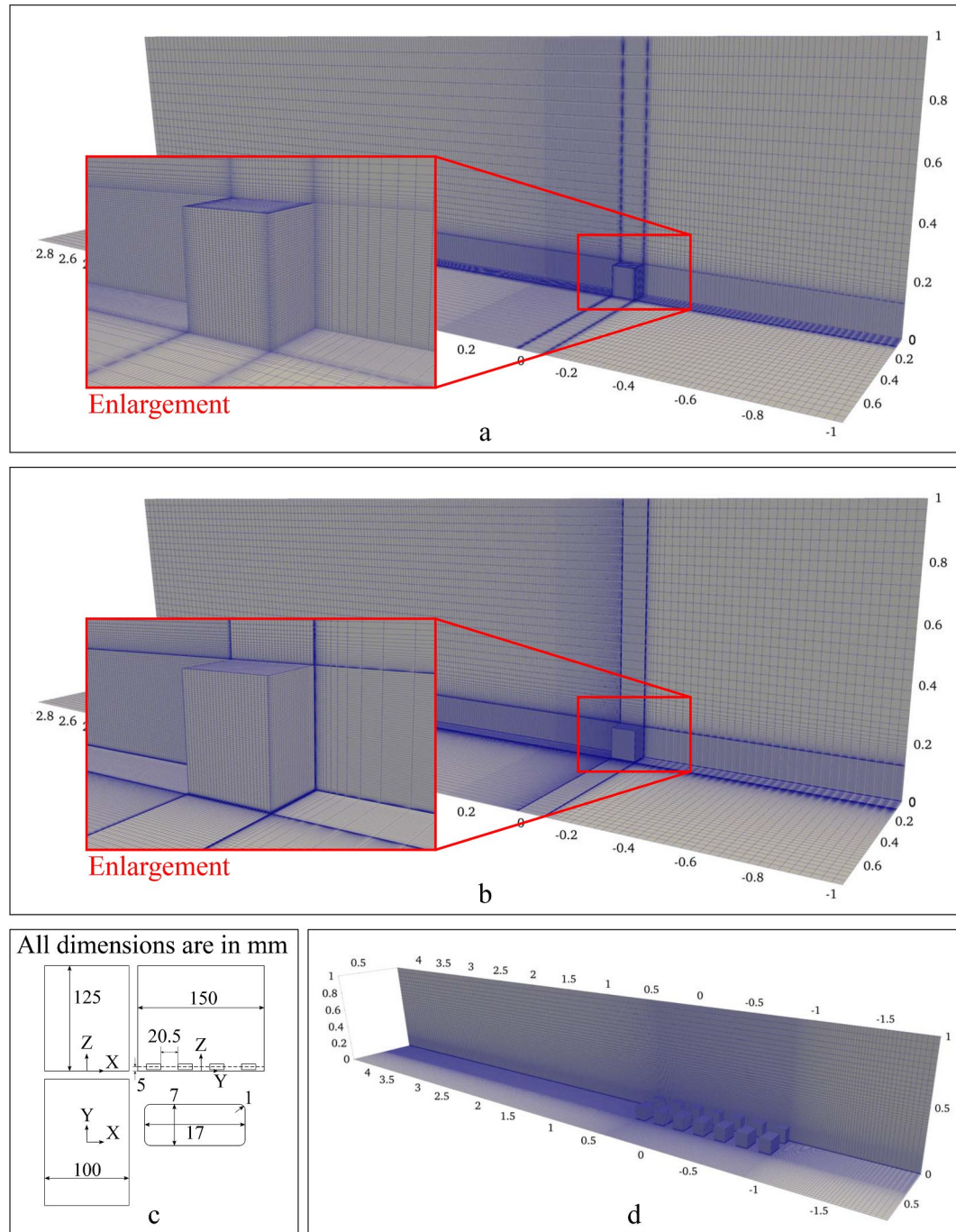


Fig. 2. (a) CEDVAL A1 test case mesh for the modified $k-\epsilon$ and modified $k-\omega$ turbulence models, (b) CEDVAL A1 test case mesh for the $SST k-\omega$ turbulence model, (c) geometry of one building, and (d) CEDVAL B1 test case mesh for the modified $k-\epsilon$ and modified $k-\omega$ turbulence models.

3.3. Full-scale empty domain

The ability of the ABL k - ϵ and k - ω models to maintain inlet profiles throughout a full-scale empty domain is tested by simulating the Silsoe test case [65]. The two-dimensional domain is 5000 m long and 500 m high. The OpenFOAM® blockMesh utility is used to create a 2D hexahedral grid, which is uniform in the streamwise direction and stretched in the cross-stream direction such that the first cell height is 1 m. The same boundary conditions used in the CEDVAL A1 case are applied and the model is tested with the four k inlet profiles given by equations (16), (19), (23), and (24). Due to the unavailability of experimental data, arbitrary parameters for the k profiles are used in equations (19) and (23), and the ability of the model to sustain the imposed inlet k profile is tested. The inlet velocity profile is obtained from the blind test data reported in Ref. [65].

3.4. CEDVAL A1 single building

This three-dimensional test case (Fig. 2a–d), which involves an open terrain with a single building, is used to validate the ability of the modified k - ϵ and modified k - ω turbulence models to transition between homogeneous and perturbed ABL flows and to compare predicted concentration values with measurements. As depicted in Fig. 2c, the model building is 100 mm long, 125 mm high, and 150 mm wide. Further, the configuration involves four sources of tracer gas representing emissions from an underground parking garage located near the building ground and on its leeward side. The origin of the Cartesian coordinate system coincides with the centroid of the bottom face of the building with the x -axis pointing in the flow direction and the z -axis upward. The geometry of the domain combined with a wind at the inlet in the x -direction creates a symmetrical wind field with respect to the $y = 0$ plane, reducing the computational domain into half the physical domain (Fig. 2a and b). The inlet and outlet boundaries are located 1 m upstream and 3 m downstream of the building's center, respectively. The OpenFOAM® blockMesh utility is used to generate the hexahedral meshes by subdividing the domain into several blocks, each having its number of divisions and expansion ratio. This reduces the grid size while ensuring accurate resolution of the near wall region.

For the k - ϵ and k - ω models, the standard forms and wall functions are used in the perturbed zone around the building, while the ABL forms and the modified wall functions (equation (40)) are applied away from the building. The mesh, which is displayed over the entire domain along with an enlarged region around the building in Fig. 2a, consists of 2.2 million elements with the maximum length of the first near wall cell not exceeding 0.7 mm. At the inlet boundary, the k profile (equation (23)) derived by Parente et al. [50] is adopted, u and ϵ are obtained from equation (16), while ω is computed using equation (30). As explained earlier, a retarding shear stress is specified at the top of the domain. Moreover, a symmetry boundary condition is enforced on the $y = 0$ plane, a slip condition is applied on the other side of the domain, and the pressure is described at the outlet. The sinusoidal blending function (equation (45)) with $\alpha = 4$ and deviation calculated according to velocity is used to blend between explicit and implicit building regions. The predicted y^+ values along the wall varied between 0.19 and 4.2 with the modified k - ϵ model and between 0.23 and 4.1 with the modified k - ω model.

Fig. 2b depicts the grid system used with the SST k - ω model over the full domain in addition to an enlarged region around the building. The mesh is finer than the one used with the modified k - ϵ and k - ω models and is composed of around six million elements with the maximum length of the first near-wall cell not exceeding 0.05 mm. The OpenFOAM® blockMesh utility is employed to generate the mesh. An automatic wall function formulation switching between low and high Reynolds number is implemented and used in the solution. For the grid used, the predicted y^+ values along the wall varied between 0.02 and 0.89. The standard treatment at walls is adopted while the same boundary conditions used

with the modified k - ϵ and k - ω models are enforced on other boundaries. Measurements of wind flow, published in the CEDVAL A1-1 database, are available within two sampling planes: $y = 0$ (603 points) and $z = 0.035m$ (643 points). For the gas source inlets, the following values are specified:

$$u = 0.024 \text{ m/s}, k = 1.5(u l)^2, \epsilon = C_\mu^{0.75} k^{1.5} / l, \omega = k^{0.5} / (C_\mu^{0.25} l), l = 0.075 l_{\text{source}}. \quad (55)$$

Concentration measurements are available in six planes: $x = 0.051m$, $y = 0$, $y = -0.06m$, $y = -0.076m$, $z = 0.01m$, and $z = 0.035m$, and are published in CEDVAL A1_5 data in dimensionless form as

$$K^* = \frac{C_{\text{measure}} u_{\text{ref}} H^2}{C_{\text{source}} Q_{\text{source}}}, \quad (56)$$

where C_{source} is the tracer gas concentration at the source, u_{ref} is the reference speed at the reference height of 0.66 m, H is the building height, and Q_{source} is the flow rate of the tracer gas at the source.

For the Lagrangian model, the following is selected: a particles' core-radius with a value of 0.5 mm to accurately describe the shape of sources, 700 particles per source per time step to ensure smoothness of the concentration field, a time step of size 0.25 ms to guarantee continuous trajectories, and a total of 6000 time steps to ensure that all particles emitted at the beginning of the simulation pass out of the domain. Moreover, the fourth order Runge-Kutta scheme is used for advection and a Schmidt number $Sc_t = 0.4$ for eddy diffusivity calculation. Perfect reflection of particles when hitting walls or symmetry planes is assumed. Furthermore, the building and ground are considered walls, the $y = 0$ plane is considered symmetry, and all other boundaries are considered open boundaries.

3.5. CEDVAL B1 array of buildings

In this test case, the flow around the array of buildings shown in Fig. 2d is simulated to validate the model over complex geometry. A total of 21 buildings are considered, each having the same geometry as the CEDVAL A1 case, distributed in three rows. The domain has the same dimensions as the wind tunnel (6.3 m long, 1 m high, and 1.5 m wide). The inlet to the domain is 1 m upstream of the first building in the array, and the outlet is 4 m downstream of the last building. Like the single building case, the blockMesh OpenFOAM® utility is used to generate hexahedral elements (Fig. 2d) by dividing the domain into several blocks. The maximum height of the near-wall cells does not exceed 0.7 mm, the resolution in the influence area of buildings is 5 mm, and the total number of cells is 3.5 million. The case is simulated using the modified k - ϵ and modified k - ω turbulence models subject to the same boundary conditions used for the single building case. Predicted y^+ along the wall varied between 0.13 and 15 for the k - ϵ model and between 0.22 and 11 for the k - ω model.

3.6. Validation metrics

The agreement between numerical predictions and measurements is quantified based on the metrics recommended by the COST guideline, which is acceptable for the considered types of simulations [66]. For that purpose, the hit rate q , FAC2 index, and geometric mean bias (MG) are used for the velocity components and the FAC2 and MG metrics for concentration. The hit rate is expressed as follows:

$$q = \frac{1}{n} \sum_{i=1}^n N_i, \quad N_i = \begin{cases} 1 & \text{if } \left| \frac{P_i - E_i}{E_i} \right| \leq \delta_r \text{ or } |P_i - E_i| \leq \delta_a, \\ 0 & \text{otherwise} \end{cases} \quad (57)$$

where q is the hit rate, n is the number of measurement points, and P_i is the predicted value of the experimental value E_i at a location i . δ_r is the relative uncertainty and it describes the allowed fractional deviation. It

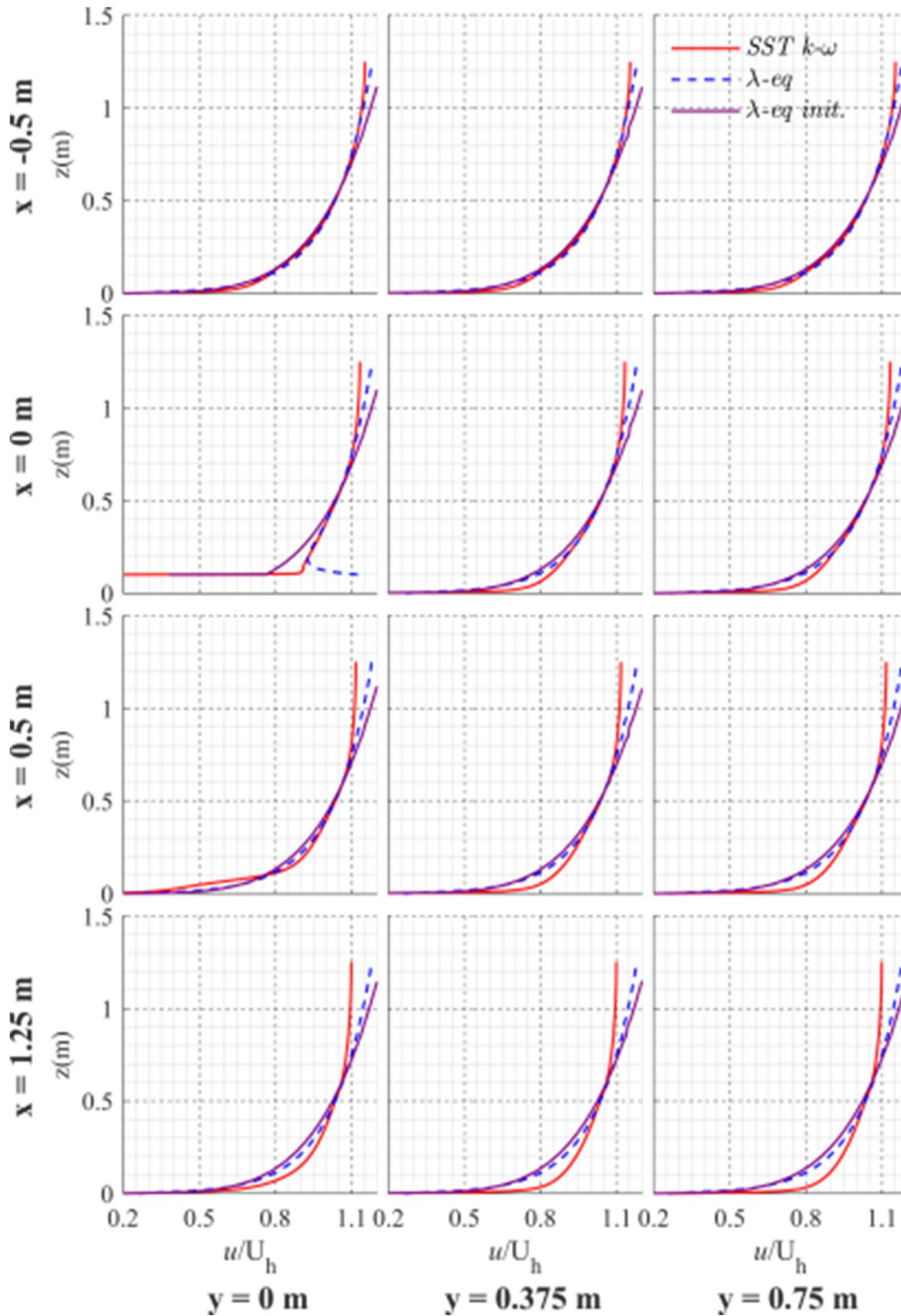


Figure 3. Comparison of prognostic ($SST k-\omega$) and diagnostic (λ -eq) normalized u-velocity profiles over the height of the domain at different x,y positions for the flow over a hemispherical obstacle.

is the sum of the allowed imprecision of simulation results and the reproducibility of wind tunnel measurements, and its value is 0.25 for the current case. δ_a is the repeatability, and it represents the allowed absolute deviation. For the velocity components, δ_a is 0.06 [67-69], whereas for the normalized concentration C , it is given as a function of its measured value by the power law $1.09c^{0.63}$ with the additional approximation $C \cong 0.083$ near C equals zero [70].

FAC2 is similar to the hit rate and represents the ratio of CFD predictions that are within a factor of two of the experimental data. It is defined as

$$FAC2 = \frac{1}{n} \sum_{i=1}^n N_i \quad N_i = \begin{cases} 1 & \text{if } 0.5 \leq \frac{P_i}{E_i} \leq 2.0 \\ 1 & \text{if } P_i \leq \delta_a \text{ and } E_i \leq \delta_a \\ 0 & \text{otherwise} \end{cases} \quad (58)$$

The geometric mean bias (MG) indicates only systematic errors and is computed based on a logarithmic scale as

$$MG = \exp(\ln \bar{E} - \ln \bar{P}), \quad E'_i = \max(\delta_a, E_i), \quad \text{and} \quad P'_i = \max(\delta_a, P_i). \quad (59)$$

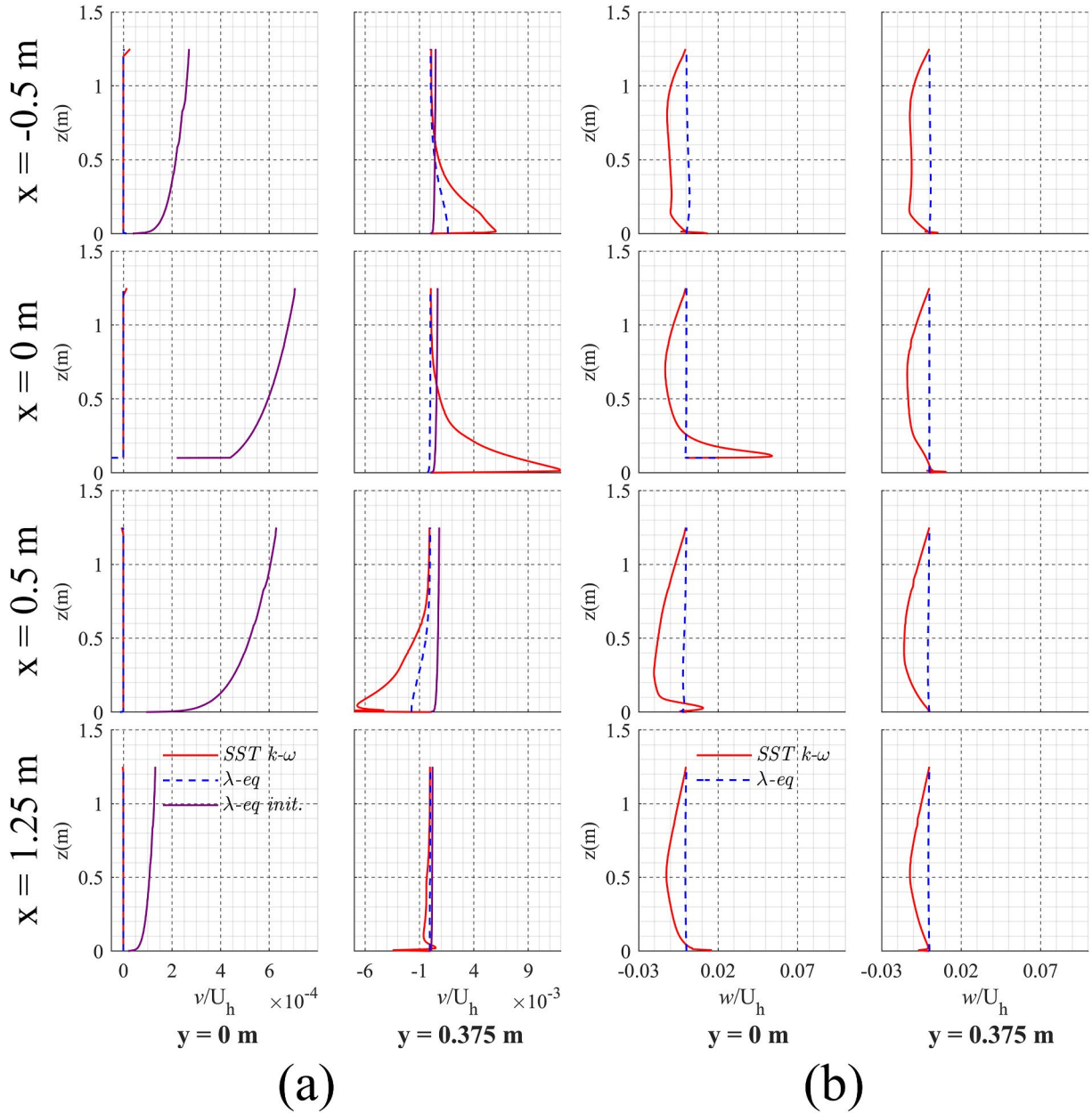


Figure 4. Comparison of prognostic (*SST k- ω*) and diagnostic(*λ -eq*) for different (a) normalized v-velocity and (b) normalized w-velocity component profiles over the height of the domain at different x,y positions for the flow over a hemispherical obstacle.

To decide whether the results generated by the model are acceptable, limits on the validation metrics should be specified. The German VDI guideline [69] requires $q \geq 0.66$ for all velocity components, while the conditions $FAC2 \geq 0.5$ and $0.7 \leq MG \leq 1.3$ have been adopted in several studies [67,71].

4. Results and discussion

4.1. Flow over a hemispherical obstacle

The velocity fields in the region close to the obstacle generated by the diagnostic and prognostic models are compared in the horizontal ($y = 0$) and vertical ($z = 0.05$ m) planes, which are shown in Fig. 1b and c, respectively. While a large recirculation region is predicted with the *SST k- ω* model, no recirculation zone is obtained with the diagnostic model. Additional comparisons of velocity component profiles over the height of the domain at four axial stations ($x = -0.5, 0, 0.5$, and 1.25 m) are

displayed in Fig. 3 and Fig. 4. Each plot is composed of three curves representing velocity-component profiles generated using the *SST k- ω* turbulence model, the initial profile constructed by interpolation from the observations using equations (1) and (2), and the final profiles obtained by adjusting the interpolated profiles using equation (4) after solving the λ -equation (i.e., equation (3)). The u-velocity profiles are depicted in Fig. 3. As shown, at each axial station, results are compared at 3 y locations ($y = 0, 0.375$, and 0.75 m). Improvements in the adjusted profiles are seen in the plots with their values lying between the *SST k- ω* and interpolated profiles. Fig. 4a and b compare, at each axial station, the v- and w-component profiles at 2 y locations ($y = 0$ and $y = 0.375$ m). The magnitudes of v and w are at least an order of magnitude lower than u. Improvements are not as clear as with the u-component. While velocity profiles generated using the prognostic model satisfy the momentum and continuity equations, adjustment to the interpolated profiles in the diagnostic approach attempts to satisfy only the continuity equation by minimizing the divergence of the velocity field over

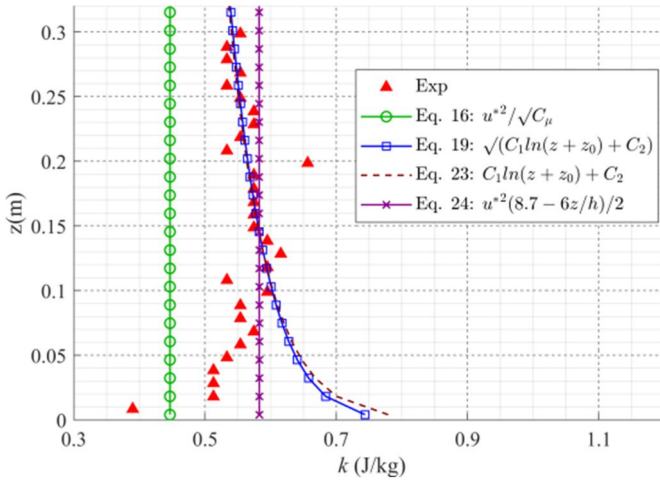


Figure 5. Comparison of experimental data with different inlet profiles for turbulent kinetic energy.

each control volume. This explains the shortcomings of the method and its inability to accurately correct the flow field, especially in domains that involve obstacles where large variation in pressure is expected. Thus, the diagnostic approach can be used to predict wind fields in simple domains away from obstacles. However, for complex geometry like flow around buildings in cities, the prognostic model is required.

4.2. CEDVAL A1 empty domain

Fig. 5 compares the various inlet profiles of the turbulent kinetic

energy with available measurements. As shown, the constant profile (equation (16)) does not describe well the experimental data while the variable profiles given by equation (19) and equation (23) show a better fit. To be used, these profiles require experimental data to calculate their fitting parameters C_1 and C_2 . Thus, it is acceptable to use the profile described by equation (24) when data is not available since this profile is close to measurements.

Fig. 6 displays profiles of turbulent kinetic energy, velocity, turbulence dissipation rate, and specific turbulence dissipation rate at the inlet and outlet sections of the empty domain used in the CEDVAL A1 test case. The performed simulations tested the use of different inlet turbulent kinetic energy profiles (Table 1a), the standard k_ϵ type (equation (38)) and modified ABL wall functions (equation (40)), and the modified $k-\epsilon$ and $k-\omega$ turbulence models. For homogenous ABL flows in empty domains, the best approach is the one that maintains the inlet profiles at the outlet section. The use of the standard wall function with the $k-\epsilon$ ABL model results in overestimation of k at the exit with a constant inlet profile for k , and underestimation in the lower part of the domain with other profiles. Similar results are obtained with the $k-\omega$ ABL model with higher underestimation and lower overestimation values. However, the use of the wall function formulation based on aerodynamic roughness resolved this issue and maintained all flow variables over the domain with all reported inlet turbulent kinetic energy profiles provided the modified wall function formulation is adopted.

4.3. Full-scale empty domain

Fig. 7 shows inlet and outlet profiles for the turbulent kinetic energy, velocity, turbulent dissipation rate, and specific dissipation rate for the full-scale empty domain case when using the modified $k-\epsilon$ and $k-\omega$

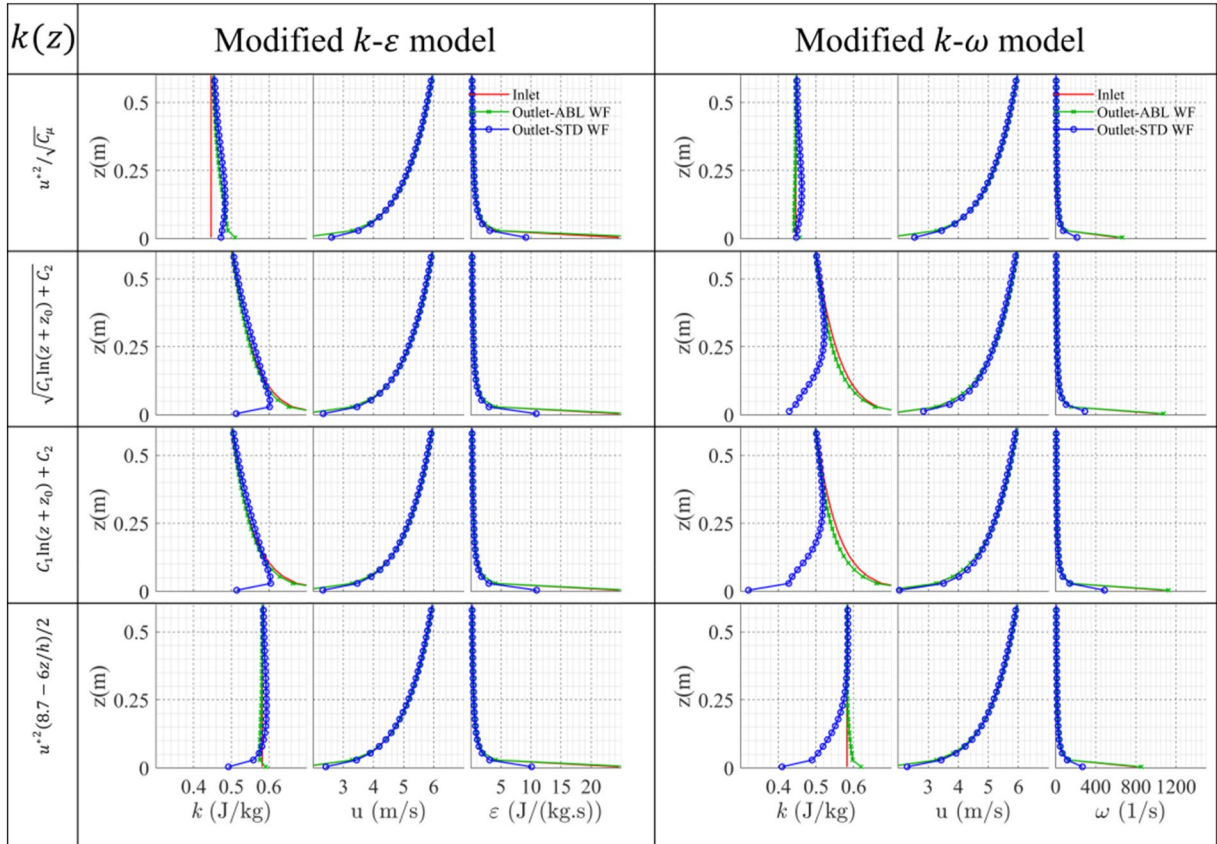


Figure 6. Comparison of inlet and outlet turbulent kinetic energy (k), x-velocity (u), turbulent dissipation rate (ϵ), and turbulent specific dissipation rate (ω) profiles for different inlet turbulent kinetic energy profiles and different wall functions for the CEDVAL A1 empty domain test case.

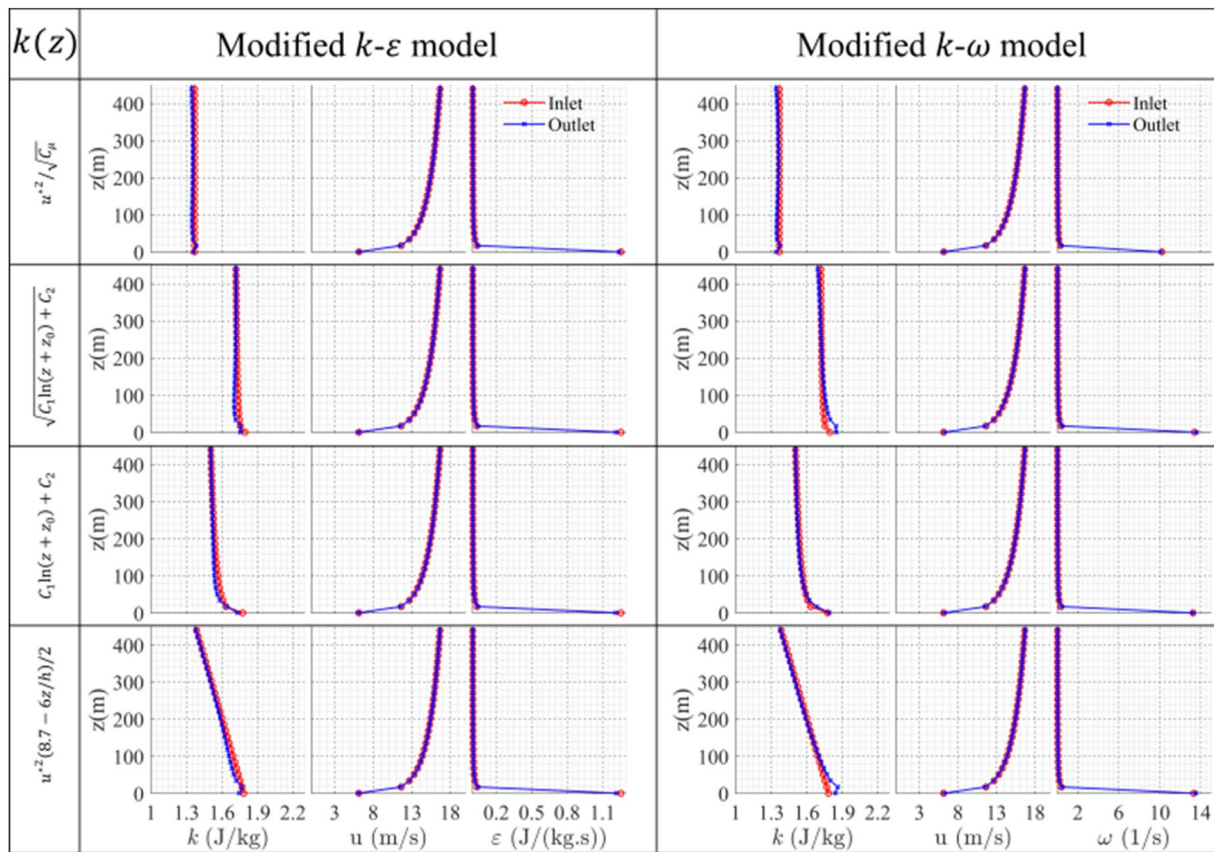


Figure 7. Comparison of inlet and outlet turbulent kinetic energy (k), x-velocity (u), turbulent dissipation rate (ϵ), and turbulent specific dissipation rate (ω) profiles for different inlet turbulent kinetic energy profiles and different wall functions for the full-scale empty domain test case.

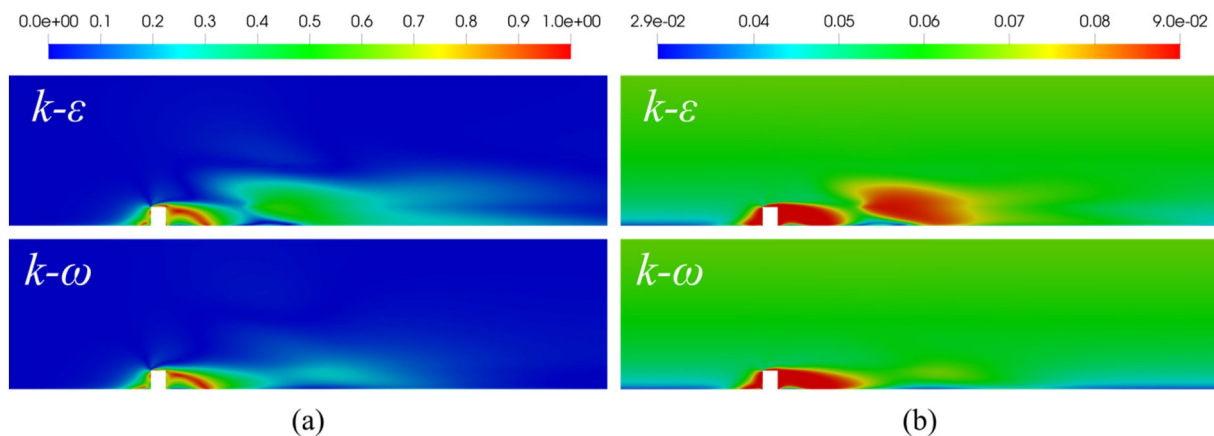


Figure 8. Contour plots for the CEDVAL A1 test case at the plane of symmetry ($y = 0$) for: (a) the blending factor δ_b when using the $k-\epsilon$ and $k-\omega$ turbulence models; (b) the coefficient C_μ when using the $k-\epsilon$ turbulence model and β^* when using the $k-\omega$ turbulence model.

models with the various inlet turbulent kinetic energy profiles. Like the CEDVAL A1 test case, results demonstrate that the modified $k-\epsilon$ and $k-\omega$ models with wall function based on aerodynamic roughness can be used to simulate ABL flows and can maintain inlet profiles in full-scale empty domains.

4.4. CEDVAL A1 single building

Contour plots of the blending factor at the symmetry plane obtained with the modified $k-\epsilon$ and modified $k-\omega$ models are displayed in Fig. 8a. The dynamic approach for determining the building influence area

results in regions of similar shapes but of sizes that depend on the turbulence model. A larger zone of influence extending further downstream is obtained with the $k-\epsilon$ model. This approach is superior to old models in which the influence area is preassigned. Moreover, Fig. 8b shows the contours of C_μ and β^* at the symmetry plane using the modified $k-\epsilon$ and modified $k-\omega$ model, respectively. Plots clearly indicate that variations in C_μ, β^* , and blending factor are similar, demonstrating the proper performance of the method.

Results depicted in Fig. 9 and Fig. 10 compare measured velocity and turbulent kinetic energy profiles with numerical predictions obtained with the modified $k-\epsilon$, modified $k-\omega$, and SST $k-\omega$ models, at different

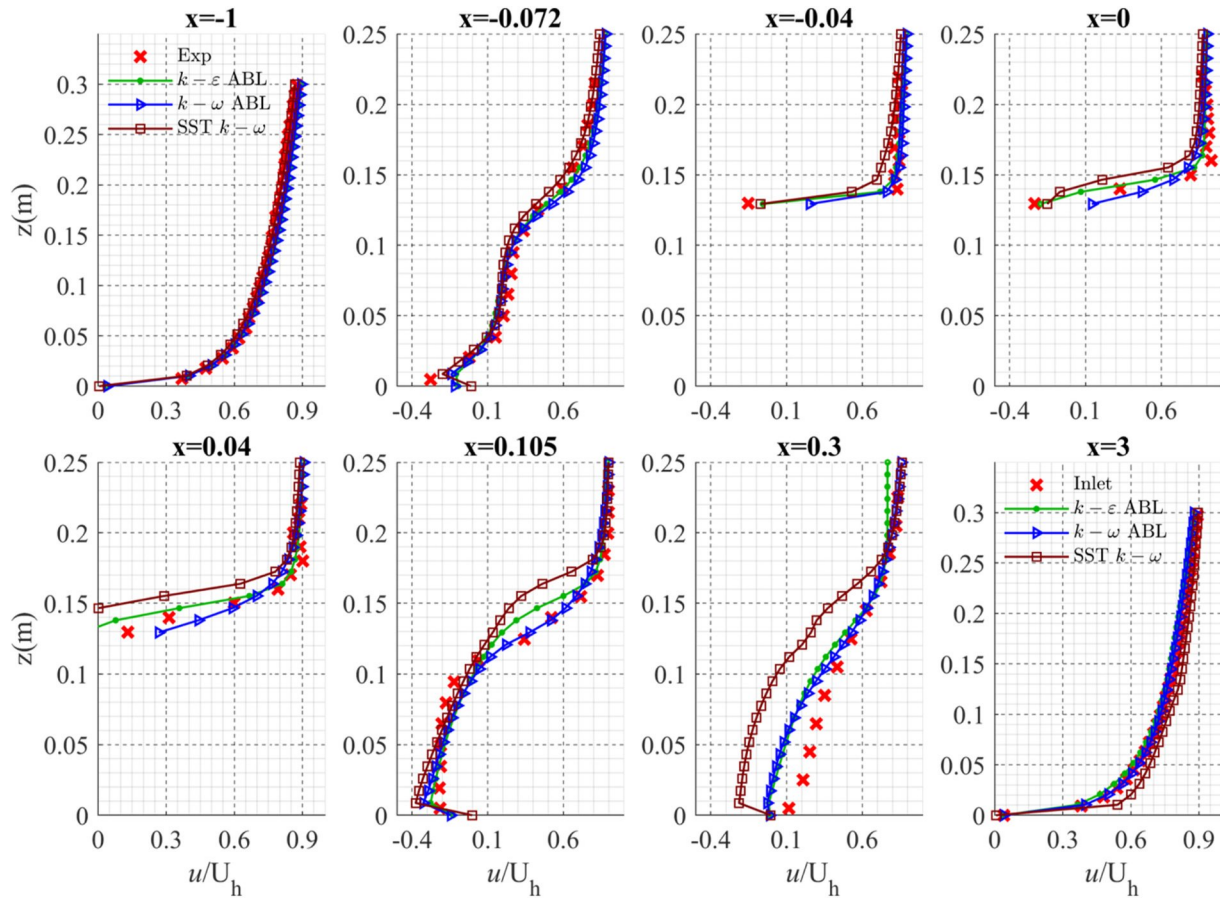


Figure 9. Comparison of predicted and measured normalized u -velocity profiles at different x positions in the $y = 0$ plane for the CEDVAL A1 case.

locations in the domain. As shown, the velocity and turbulent kinetic energy profiles using the modified $k-\epsilon$ and $k-\omega$ models are closer to the experimental measurements than the ones obtained with the $SST k-\omega$ model. This demonstrates the advantage of using the $k-\epsilon$ and $k-\omega$ models with nonuniform k profiles at the inlet, wall functions based on aerodynamic roughness, and automatic blending functions for detecting the building influence area.

Table 2a presents the validation metrics of the normalized velocity components predicted with the various turbulence models. As shown, the three models successfully simulate the flow with the modified $k-\omega$ model having the best metrics, followed by the modified $k-\epsilon$ model, which is slightly better than the standard $k-\omega$ SST model. These results confirm the conclusions drawn from the velocity and turbulent kinetic energy profiles.

Although the modified $k-\epsilon$ and modified $k-\omega$ models are better at simulating the flow, the $SST k-\omega$ model is better at simulating gas dispersion. Results displayed in Table 2b indicate that the Eulerian concentration field generated with the $k-\epsilon$ model failed in the FAC2 and MG validation tests while the $k-\omega$ barely succeeded in the FAC2 test, but also failed in the MG validation test. On the other hand, the Eulerian and Lagrangian concentration fields obtained with the $SST k-\omega$ model meet the criteria for both FAC2 and MG metrics. However, while the Eulerian and Lagrangian FAC2 values are close, the MG metric shows a positive bias when using the Eulerian model and a negative bias when using the Lagrangian model. This behavior will be clarified in the discussions to follow.

The use of an automatic wall function formulation with the $SST k-\omega$ model allows switching between a high and a low Reynolds number model based on the mesh size. Moreover, the mesh near walls for the $SST k-\omega$ model is finer than that used with the $k-\epsilon$ and $k-\omega$ models. This gives an advantage for the $SST k-\omega$ model to better predict the velocity field in

this region even though the modified $k-\epsilon$ and modified $k-\omega$ models show better validation metrics in the domain. Further, because the gas sources emanate from the bottom of the building, they are largely affected by the velocity field near the walls. This explains the success of the $SST k-\omega$ model in predicting the concentration field, unlike other models.

Maps of normalized u -velocity (u/U_h) and turbulent kinetic energy (k/U_h^2) in the $y = 0$ plane are displayed in Fig. 11a and Fig. 11b, respectively.

Although results shown in Fig. 11(a) and (b) are similar for the various models, the velocity vector fields shown in Fig. 12 are slightly different. The differences in this critical zone affect the dispersion of gas, which is emitted from that region. Two parameters are important when comparing velocity vectors in the building influence area, which are the wake recirculation length (L_W) and the rooftop reattachment length (L_R). The rooftop reattachment length appears in the $y = 0$ plane (i.e., the xz -plane) in Fig. 12, and represents the distance traveled by the fluid before reattaching to the flow over the building. Moreover, L_W signifies the length of recirculation on the leeward side of the building. The different values of (L_R) and (L_W) shown in Fig. 12 lead to significant effects on the way the gas is advected. Further, predictions indicate that the air will not reattach on the top and the side of the building except for the case when using the modified $k-\omega$ model, which predicted a reattachment in the xz -plane (when using the modified $k-\omega$ model, L_R in xz -plane is shorter than the building size in the x direction). Besides, the modified $k-\epsilon$ and $k-\omega$ models show similar sizes of wake recirculation lengths, and these lengths are close to the length predicted by the $SST k-\omega$ model, unlike the results for the standard $k-\epsilon$ turbulence model reported in Ref. [67]. Moreover, all the models predicted the secondary recirculation at the corner behind the building, shown as enlarged regions in Fig. 11, with the one predicted by the $SST k-\omega$ model being the

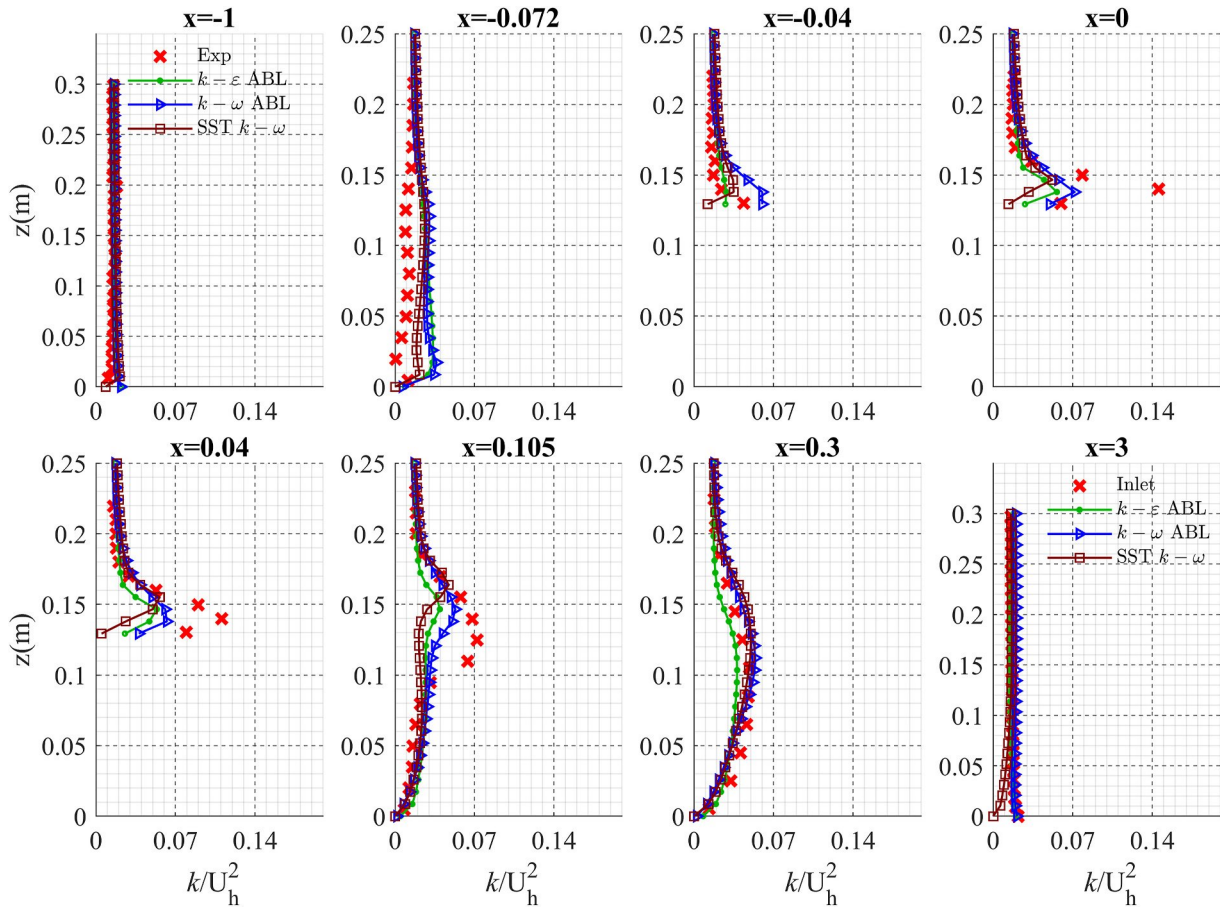


Figure 10. Comparison of predicted and measured normalized turbulent kinetic energy profiles at different x positions in the $y = 0$ plane for the CEDVAL A1 case.

Table 2a

Summary of the $k-\epsilon$ and $k-\omega$ models validation metrics for the single building case.

Model	$k-\epsilon$ model			$k-\omega$ model			SST $k-\omega$ model		
	u/U_{ref}	v/U_{ref}	w/U_{ref}	u/U_{ref}	v/U_{ref}	w/U_{ref}	u/U_{ref}	v/U_{ref}	w/U_{ref}
Hit rate ≥ 0.66	0.66	0.76	0.93	0.71	0.87	0.94	0.68	0.79	0.86
FAC2 ≥ 0.5	0.84	0.91	0.98	0.87	0.96	0.98	0.74	0.90	0.91
MG (0.7 – 1.3)	1.20	0.92	0.97	1.19	0.95	1.00	1.23	0.92	0.92

Table 2b

Summary of validation metrics for normalized concentration in the single building case.

Flow Model	$k-\epsilon$	$k-\omega$	SST $k-\omega$	SST $k-\omega$
	Eulerian	Eulerian	Eulerian	Lagrangian
FAC2 ≥ 0.5	0.40	0.50	0.68	0.67
MG (0.7 – 1.3)	0.40	0.50	1.19	0.88

clearest. These secondary recirculation zones are very important for modeling the dispersion in the considered case because the gas emanates from that region.

These effects are noticed in Fig. 13a–e where maps of the normalized concentration fields predicted by the different models are presented and compared with measurements. The Eulerian concentration field obtained with the $k-\epsilon$ model (Fig. 13a) indicates that a large portion of the pollutant gas recirculates to the windward side of the building before continuing in the flow direction. Moreover, the $k-\omega$ Eulerian field (Fig. 13b) looks more diffusive with concentration not following strictly the wind field but spreading around it. On the other hand, the Eulerian

concentration field obtained with the SST $k-\omega$ model (Fig. 13c) compares well with the experimental field presented in Fig. 13e [43]. Finally, the shape of the concentration field predicted using the SST $k-\omega$ with the Lagrangian concentration model (Fig. 13d) is not closer to the experimental field than its Eulerian counterpart despite passing all validation metrics.

The Lagrangian model for advection-diffusion of the concentration field involves simulating advection and diffusion in two separate steps. The particles are injected into the field with a core radius of 0.5 mm, which is a fraction of the grid cell size (nearly one-third). To ensure overlap, which is required for smoothness of the concentration field [72], 1400 particles were uniformly distributed over the inlet face, resulting in an overlap of 1.213 $\left(overlap = \sigma / \left(\frac{area}{particles}\right)^{0.5}\right)$. To further promote overlap, the particles are injected every 0.25 ms. Given that the inlet speed is 0.024 m/s, they travel 6×10^{-6} m per time step, ensuring an overlap of nearly 83. The overlap condition is however not guaranteed as the particles are advected along the path lines causing them to get closer to each other in some regions and farther apart in other regions. Since diffusion is modeled by core spreading, the core size of a particle

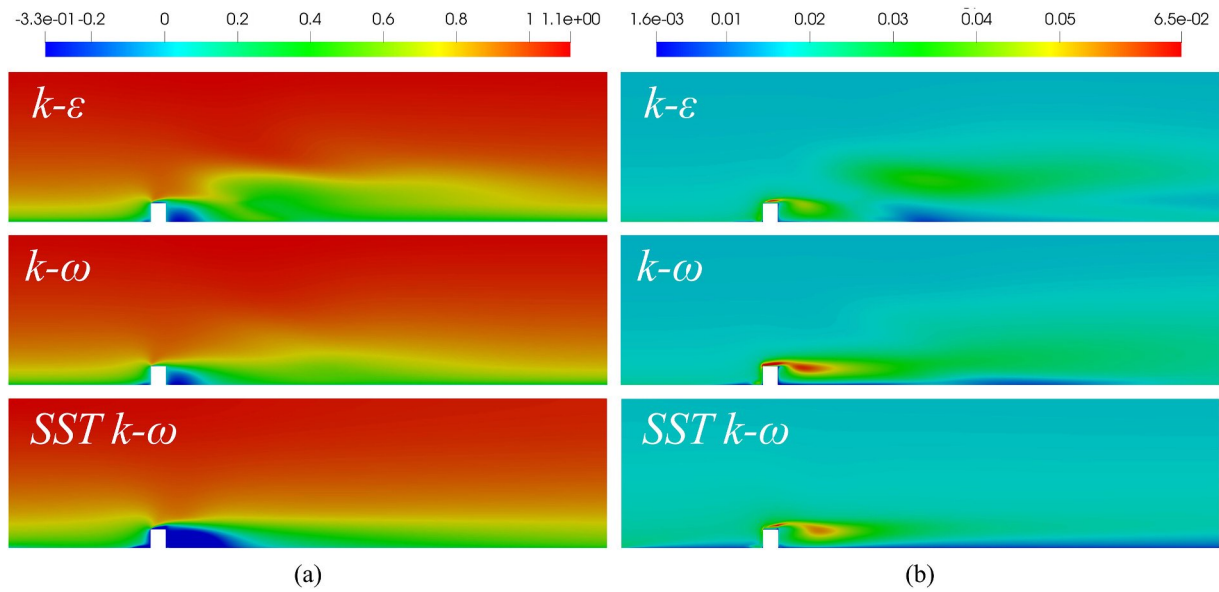


Figure 11. Contour plots of (a) the normalized u -velocity (u/U_h) and (b) the normalized turbulent kinetic energy (k/U_h^2) obtained using the different turbulence models in the CEDVAL A1 test case.

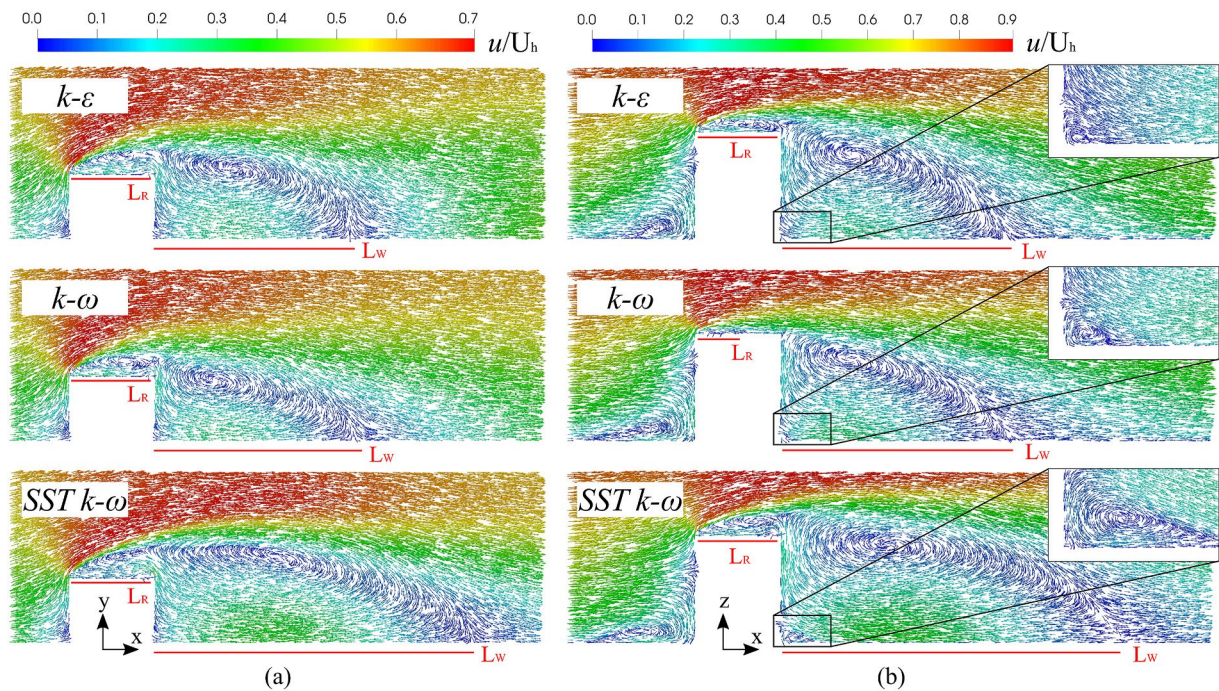


Figure 12. Normalized velocity vector plots for the CEDVAL A1 test case obtained using the modified $k-\epsilon$, the modified $k-\omega$, and the $SST k-\omega$ turbulence models in (a) $z = 0.035$ m and (b) $y = 0$ planes.

grows as $\sqrt{D * t}$ (D depends on the turbulent viscosity of the flow), implying that the core size of a particle in the domain is decided by the level of turbulence and how long it has been in the domain since injection. As such particles that are trapped in recirculation zones may grow to much larger values (computed as 0.26 m) than those that follow a straight line along the inlet velocity for which their residence time (domain length/wind speed) is much shorter. As the core size of a particle grows, the particles' overlap increases, which improves the smoothness of the solution. The downside is that this introduces errors in the advection step [73]. These observations are reflected in Fig. 14 (a) and (b) which show respectively the distribution of particles, colored by their core size, for cases of flow over one building and flow over an array

of buildings. The corresponding concentration distribution for the case of a single building is shown in Fig. 13 (d). When compared to the concentration distributions predicted by the Eulerian concentration solvers in Fig. 13 (a), (b), and (c), it can be seen that the Lagrangian method captures the small-scale fractures of the concentration field in the immediate vicinity of the building with little artificial diffusion while it suffers from considerable artificial diffusion in the far field and near the symmetry plane, which is caused by the particle with oversized core radii. Errors caused by oversized particles can be significantly reduced, while meeting the overlap conditions, by the implementation of diffusion schemes that involve particle splitting such as the redistribution scheme [74].

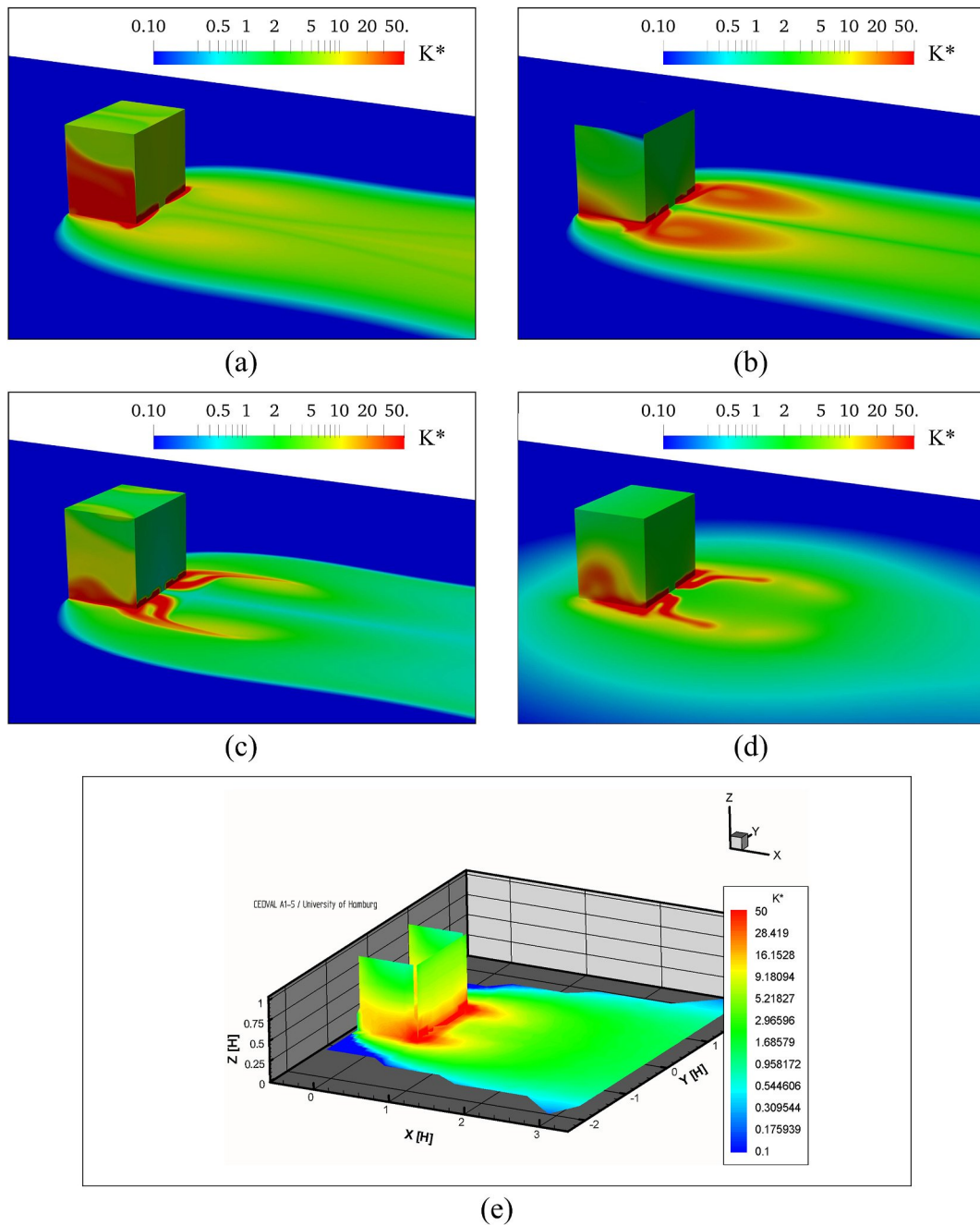


Figure 13. Normalized concentration for the CEDVAL A1 test case when using (a) the modified $k-\epsilon$ model, (b) the modified $k-\omega$ model, (c) and the $SST\ k-\omega$ model with the Eulerian concentration solver; (d) the $SST\ k-\omega$ model with the Lagrangian concentration solver, and (e) the experimental results from CEDVAL A1-5 data set [43].

With the Lagrangian model, the concentration field follows strictly the particles' trajectories with higher concentration levels in the building influence zone at the injection location. The fact that the available experimental data of the CEDVAL A1_5 test case are mostly near the building, explains why the Lagrangian MG metric has a negative bias, which indicates that predictions are higher than measurements. The Lagrangian particle trajectories shown in Fig. 14a follow the Eulerian streamlines shown in Fig. 14c because the density of the particles is equal to the density of the air and the sizes of the gas particles are too small. This comparison demonstrates the accuracy of the Lagrangian method.

4.5. CEDVAL B1 array of buildings

The validation metrics for the velocity components predicted using the modified $k-\epsilon$ and $k-\omega$ model with aerodynamic rough wall functions for the array of buildings case are reported in Table 2c.

Results indicate success in the hit rate q , FAC2, and MG validation metrics for both models. Comparisons at different locations in the domain of the predicted normalized velocity and turbulent kinetic energy profiles with measured profiles are displayed in Fig. 15a and Fig. 15b, respectively. As shown, numerical predictions agree with experimental data, which is consistent with the validation metrics. Moreover, the contours of the blending factor displayed in Fig. 15c indicate that the location of the detected building influence zone is as

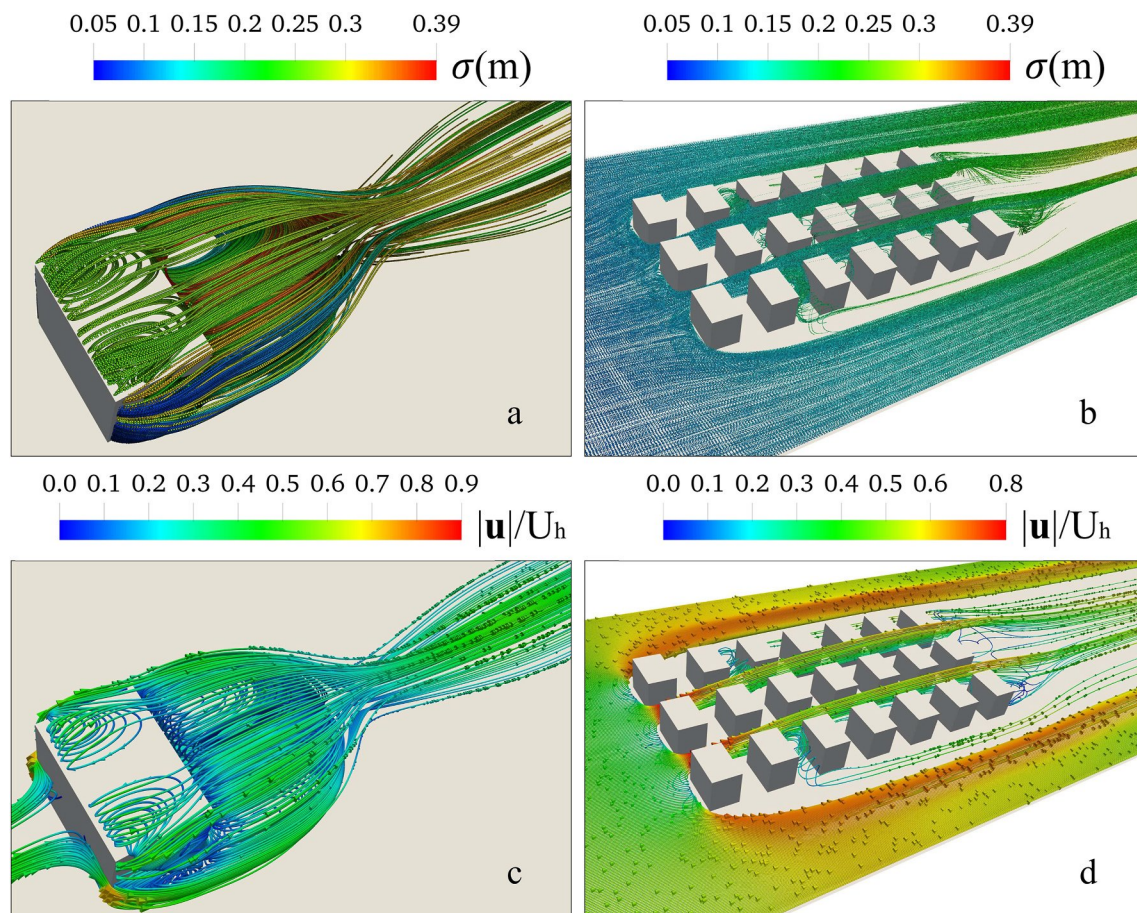


Figure 14. Lagrangian representation of the distribution of particles for (a) one building domain and (b) for an array of buildings; Streamlines using the Eulerian approach for (c) one building, and (d) for an array of buildings.

Table 2c

Summary of the $k-\epsilon$ and $k-\omega$ models validation metrics for the array of buildings case.

Model	$k-\epsilon$ model			$k-\omega$ model		
	u/u_{ref}	v/u_{ref}	w/u_{ref}	u/u_{ref}	v/u_{ref}	w/u_{ref}
Hit rate ≥ 0.66	0.81	0.67	0.71	0.67	0.73	0.77
FAC2 ≥ 0.5	0.93	0.99	0.75	0.89	1.00	0.77
MG (0.7 – 1.3)	1.0882	1.00	1.18	1.03	0.99	1.16

expected. From this simulation, it can be concluded that the modified turbulence model can be confidently used to predict ABL flows in domains with several buildings. As for the one building case, streamlines predicted using the Eulerian approach (Fig. 14b) and particle trajectories obtained using the Lagrangian approach (Fig. 14d) are consistent. This demonstrates the capability of the Lagrangian model to predict particle trajectories.

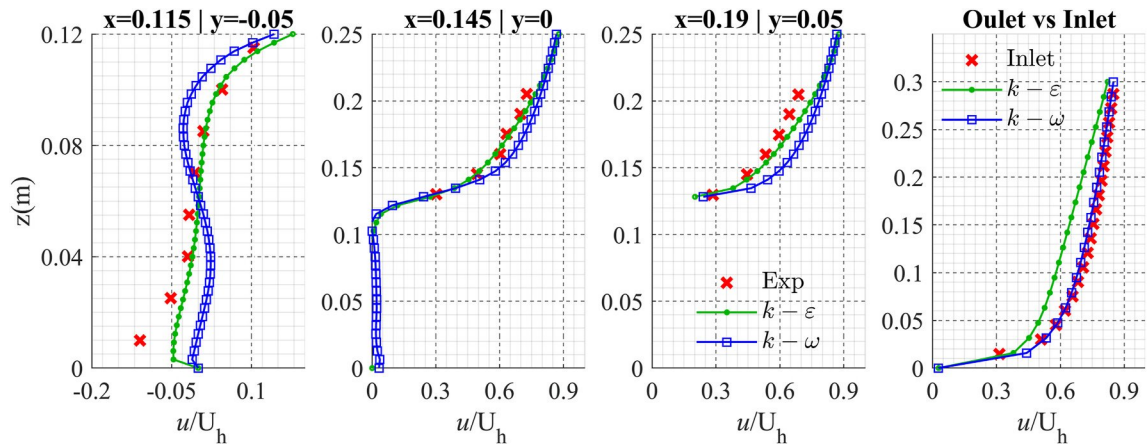
5. Closing Remarks

In this study, a modified $k-\omega$ turbulence model for the prediction of atmospheric boundary layer flows is developed. The modified model is consistent with wall functions and inlet conditions and helps maintain the inflow conditions through an empty domain for any inlet turbulent kinetic energy profile. The new model is integrated into a numerical tool embodying the diagnostic and prognostic approaches for simulating wind flow and pollutant concentration in urban areas. The diagnostic model is based on solving an anisotropic diffusion equation for the Lagrange multiplier and can be used to obtain quick but approximate

results in simple open domains. For complex domains, the prognostic approach that is based on solving the Navier-Stokes equations with a turbulence model should be employed. To overcome the limitations of the sand-grain rough wall functions and to have more flexibility in mesh generation, wall functions based on aerodynamic roughness were adopted. By simulating the flow in an empty domain using the standard and the modified wall functions for different turbulent kinetic energy inlet profiles, it was found that the wall function formulation based on aerodynamic roughness can maintain inlet conditions when using the $k-\epsilon$ or $k-\omega$ ABL models. Moreover, numerical simulations for wind field and particles dispersion were performed for the case of a single building using the modified $k-\epsilon$, modified $k-\omega$, and SST $k-\omega$ model, and for the wind field for the case of an array of buildings using the modified $k-\epsilon$ and modified $k-\omega$ models. Simulation results were rigorously tested by statistical validation metrics using extensive experimental data. Accordingly, it was found that flow field predictions obtained with the modified $k-\epsilon$ and modified $k-\omega$ models were more accurate than the ones generated with the SST $k-\omega$ model. However, the opposite is true with the predictions of pollutant dispersion originating from a pollutant source on the leeward side of the building. By comparing the Lagrangian and Eulerian models for simulating gas distribution it can be concluded that the use of the particle method with core-radius spreading for diffusion is a good choice if the velocity gradient in the volume that a particle represents is not high.

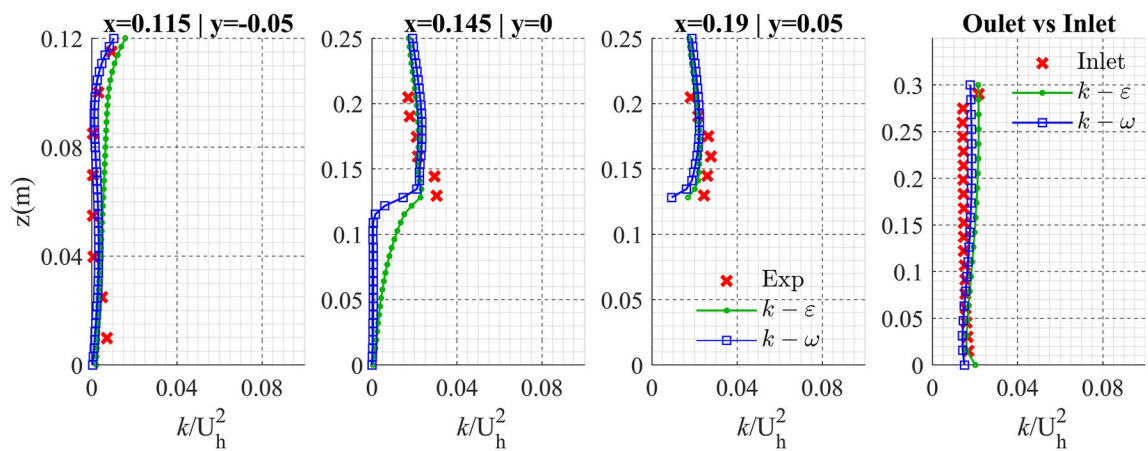
CRediT authorship contribution statement

A. Saleh: Software, Investigation, Formal analysis, Data curation, Conceptualization. **I. Lakkis:** Supervision, Project administration,



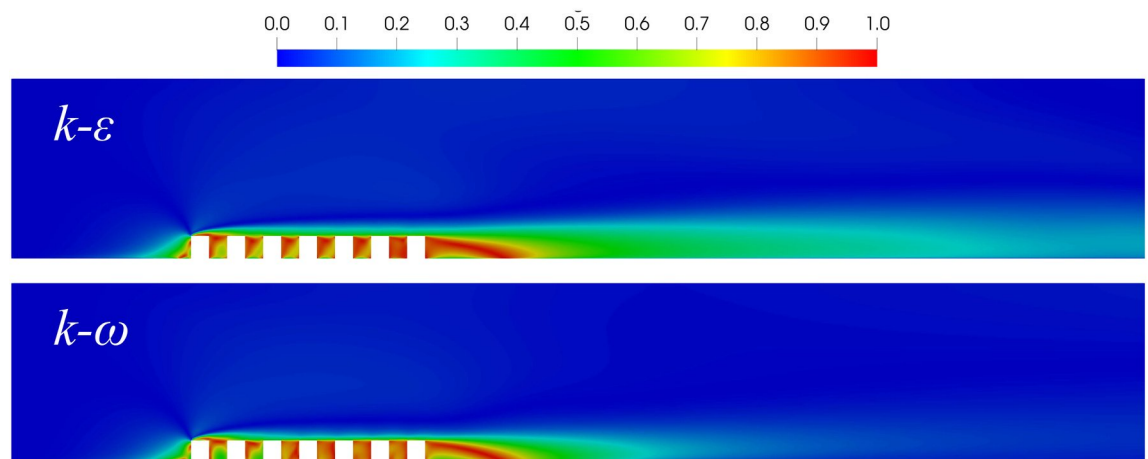
(a)

all dimensions are in meter



(b)

all dimensions are in meter



(c)

Figure 15. Comparison of predicted and measured normalized (a) u-velocity profiles and (b) turbulent kinetic energy profiles at different x,y locations over the height of the domain for the CEDVAL B1 test case; (c) Blending factor for CEDVAL B1 test case obtained with the modified $k-\epsilon$ and $k-\omega$ turbulence models.

Methodology, Conceptualization. **F. Moukalled:** Writing – original draft, Supervision, Software, Project administration, Methodology, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence

the work reported in this paper.

Data availability

Data will be made available on request.

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